

**Subject: Discrete Mathematical Structures**

**Subject Code: BCS405A**

**Semester: 4th**

**Scheme: 2022**

## **Module-3: Relations and Functions**

- ➔ **Cartesian Products, Relations and Functions**
- ➔ **Types of Functions**
- ➔ **Pigeonhole Principle**
- ➔ **Composite functions**
- ➔ **Invertible Functions**
- ➔ **Properties of Functions**
- ➔ **Zero-one matrices and Directed graphs**
- ➔ **Properties of Relations**
- ➔ **Equivalence relation**
- ➔ **Partial orders**
- ➔ **External elements in Posets**

### Cartesian Product of sets:

Let  $A$  and  $B$  be 2 sets. Then the set of all ordered pairs  $(a, b)$ , where  $a \in A$  and  $b \in B$ , is called Cartesian Product (or) Cross Product (or) Product set of  $A$  and  $B$  and is denoted by  $A \times B$ .

Thus  $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$ .

Note: 1)  $A \times B$  is not same as  $B \times A$ , because

$$B \times A = \{(b, a) \mid b \in B \text{ and } a \in A\}.$$

$\therefore (a, b) \neq (b, a)$  in general.

2) If  $(a, b)$  and  $(c, d)$  are ordered pairs, then  $(a, b) = (c, d)$  if and only if  $a = c, b = d$ .

3) If  $A$  and  $B$  are finite sets with  $|A| = m, |B| = n$ , then

$$|A \times B| = mn = |A||B|.$$

4)  $|P(A \times B)| = 2^{mn}$  i.e.  $A \times B$  has  $2^{mn}$  subsets where  $|A| = m, |B| = n$ . For ex: if  $A$  has 5 elements,  $B$  has 6 elements then  $A \times B$  has  $2^{5 \times 6} = 2^{30}$  subsets.

### Problems:

1) Let  $A = \{1, 2, 3, 4\}$      $B = \{3, 4, 5, 6\}$      $C = \{2, 4, 6\}$ . Find  $A \times B, B \times A, A \times (B \cup C), (A \cap B) \times C, (A \times B) \cap (B \times C), (A \times B) - (B \times C)$ .

Soln:-  $A \times B = \{(1, 3) (1, 4) (1, 5) (1, 6) (2, 3) (2, 4) (2, 5) (2, 6) (3, 3) (3, 4) (3, 5) (3, 6) (4, 3) (4, 4) (4, 5) (4, 6)\}$

$B \times A = \{(3, 1) (3, 2) (3, 3) (3, 4) (4, 1) (4, 2) (4, 3) (4, 4) (5, 1) (5, 2) (5, 3) (5, 4) (6, 1) (6, 2) (6, 3) (6, 4)\}$

$B \cup C = \{2, 3, 4, 5, 6\}$ .

$A \times (B \cup C) = \{(1, 2) (1, 3) (1, 4) (1, 5) (1, 6) (2, 2) (2, 3) (2, 4) (2, 5) (2, 6) (3, 2) (3, 3) (3, 4) (3, 5) (3, 6) (4, 2) (4, 3) (4, 4) (4, 5) (4, 6)\}$

$$(A \cap B) = \{3, 4\} \quad , \quad C = \{2, 4, 6\}.$$

$$(A \cap B) \times C = \{(3, 2) (3, 4) (3, 6) (4, 2) (4, 4) (4, 6)\}.$$

$$(B \times C) = \{(3, 2) (3, 4) (3, 6) (4, 2) (4, 4) (4, 6) (5, 2) (5, 4) (5, 6) (6, 2) (6, 4) (6, 6)\}.$$

$$\therefore (A \times B) \cap (B \times C) = \{(3, 4) (3, 6) (4, 4) (4, 6)\}.$$

$$(A \times B) - (B \times C) = \{(1, 3) (1, 4) (1, 5) (1, 6) (2, 3) (2, 4) (2, 5) (2, 6) (3, 3) (3, 5) (4, 3) (4, 5)\}.$$

elements in  $(A \times B)$   
but not in  $(B \times C)$ .

2) For any non-empty sets  $A, B, C$ , prove the following results:

- i)  $A \times (B \cup C) = (A \times B) \cup (A \times C)$
- ii)  $(A \cup B) \times C = (A \times C) \cup (B \times C)$
- iii)  $A \times (B \cap C) = (A \times B) \cap (A \times C)$
- iv)  $(A \cap B) \times C = (A \times C) \cap (B \times C)$
- v)  $A \times (B - C) = (A \times B) - (A \times C)$

Soln:- For any ordered pair  $(x, y)$ , we have

$$\begin{aligned} \text{i) } (x, y) \in \{A \times (B \cup C)\} &\Leftrightarrow x \in A \text{ and } y \in B \cup C \\ &\Leftrightarrow x \in A \text{ and } y \in B \text{ or } y \in C. \\ &\Leftrightarrow x \in A \text{ and } y \in B, \text{ or } x \in A \text{ and } y \in C. \\ &\Leftrightarrow (x, y) \in A \times B \text{ or } (x, y) \in A \times C \\ &\Leftrightarrow (x, y) \in \{(A \times B) \cup (A \times C)\}. \end{aligned}$$

$$\text{Thus } A \times (B \cup C) = (A \times B) \cup (A \times C).$$

$$\begin{aligned} \text{ii) } (x, y) \in \{(A \cup B) \times C\} &\Leftrightarrow x \in A \cup B, \text{ and } y \in C. \\ &\Leftrightarrow x \in A \text{ or } x \in B \text{ and } y \in C \\ &\Leftrightarrow x \in A \text{ and } y \in C \text{ or } x \in B \text{ and } y \in C. \\ &\Leftrightarrow (x, y) \in A \times C \text{ or } (x, y) \in B \times C. \\ &\Leftrightarrow (x, y) \in \{(A \times C) \cup (B \times C)\}. \end{aligned}$$

$$\text{Thus } (A \cup B) \times C = (A \times C) \cup (B \times C).$$

$$\text{iii) } (x, y) \in \{A \times (B \cap C)\} \Leftrightarrow x \in A \text{ and } y \in B \cap C.$$

$$\Leftrightarrow x \in A \text{ and } y \in B \text{ and } y \in C.$$

$$\Leftrightarrow x \in A \text{ and } y \in B, \text{ and } x \in A \text{ and } y \in C.$$

$$\Leftrightarrow (x, y) \in A \times B \text{ and } (x, y) \in A \times C.$$

$$\Leftrightarrow (x, y) \in \{(A \times B) \cap (A \times C)\}$$

$$\text{Thus } A \times (B \cap C) = (A \times B) \cap (A \times C).$$

$$\text{iv) } (x, y) \in (A \cap B) \times C \Leftrightarrow x \in A \cap B \text{ and } y \in C.$$

$$\Leftrightarrow x \in A \text{ and } x \in B, \text{ and } y \in C.$$

$$\Leftrightarrow x \in A \text{ and } y \in C \text{ and } x \in B \text{ and } y \in C$$

$$\Leftrightarrow (x, y) \in (A \times C) \text{ and } (x, y) \in (B \times C)$$

$$\Leftrightarrow (x, y) \in \{(A \times C) \cap (B \times C)\}.$$

$$\text{Thus } (A \cap B) \times C = (A \times C) \cap (B \times C).$$

$$\text{v) } A \times (B - C) = (A \times B) - (A \times C).$$

$$(x, y) \in \{A \times (B - C)\} \Leftrightarrow x \in A \text{ and } y \in B - C.$$

$$\Leftrightarrow x \in A \text{ and } y \in B, \text{ and } y \notin C.$$

$$\Leftrightarrow x \in A \text{ and } y \in B, \text{ and } x \in A \text{ and } y \notin C$$

$$\Leftrightarrow (x, y) \in (A \times B) \text{ and } (x, y) \notin (A \times C)$$

$$\Leftrightarrow (x, y) \in \{(A \times B) - (A \times C)\}.$$

$$\text{Thus } A \times (B - C) = (A \times B) - (A \times C).$$

$$\text{3) suppose } A, B, C \subseteq \mathbb{R} \times \mathbb{R} \text{ with } A = \{(x, y) \mid y = 5x - 1\},$$

$$B = \{(x, y) \mid y = 6x\}, \quad C = \{(x, y) \mid 3x - y = -7\}.$$

$$\text{Find i) } A \cap B \quad \text{ii) } B \cap C \quad \text{iii) } \overline{A \cup C} \quad \text{iv) } \overline{B \cup C}.$$

$$\text{soln:- i) } (x, y) \in A \cap B \Leftrightarrow (x, y) \in A \text{ and } (x, y) \in B.$$

$$\Leftrightarrow y = 5x - 1 \text{ and } y = 6x.$$

$$\Leftrightarrow 5x - 1 = y = 6x.$$

$$\Leftrightarrow x = -1, y = -6.$$

$$\therefore A \cap B = \{(-1, -6)\}.$$

$$ii) (x, y) \in B \cap C \Leftrightarrow (x, y) \in B \text{ and } (x, y) \in C.$$

$$\Leftrightarrow y = 6x \text{ and } 3x - y = -7.$$

$$\Leftrightarrow y = 6x \text{ and } y = 3x + 7$$

$$\Leftrightarrow 6x = y = 3x + 7$$

$$\Leftrightarrow 3x = 7 \quad \underline{\text{ie}} \quad x = 7/3$$

which is not possible, because  $x \in \mathbb{Z}$ .

$$\text{Thus } (B \cap C) = \phi.$$

$$iii) \text{ we have } \overline{A \cup C} = \overline{A \cap C}$$

$$\Rightarrow \overline{A \cup C} = \overline{A \cap C} = A \cap C.$$

$$(x, y) \in A \cap C \Leftrightarrow (x, y) \in A \text{ and } (x, y) \in C.$$

$$\Leftrightarrow y = 5x - 1 \text{ and } 3x - y = -7.$$

$$\Leftrightarrow y = 5x - 1 \text{ and } y = 3x + 7$$

$$\Leftrightarrow 5x - 1 = y = 3x + 7$$

$$\Leftrightarrow x = 4, y = 19.$$

$$\text{Thus } \overline{A \cup C} = A \cap C = \{(4, 19)\}.$$

$$iv) \text{ we have } \overline{B \cup C} = \overline{B \cap C}.$$

$$\text{from (ii), } B \cap C = \phi.$$

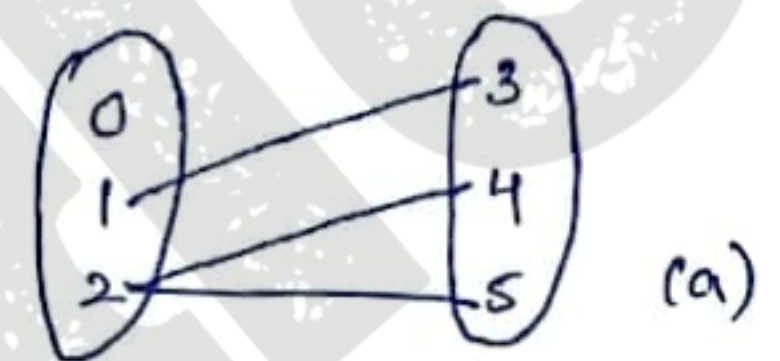
$$\therefore \overline{B \cap C} = \overline{B \cup C} = \mathbb{Z} \times \mathbb{Z} \text{ (universal set).}$$

Relations :- Let  $A$  and  $B$  be 2 sets. Then a subset of  $A \times B$  is called a binary relation or just a relation from  $A$  to  $B$ . Thus (if  $R$  is a relation from  $A$  to  $B$ , then  $R$  is a set of ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ ), and conversely if  $R$  is a set of ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ , then  $R$  is a relation from  $A$  to  $B$ .

If  $(a, b) \in R$ , we say that " $a$  is related to  $b$  by  $R$ ". This is denoted by  $aRb$ . If  $R$  is a relation from  $A$  to  $A$  i.e.  $R$  is a subset of  $A \times A$ , we say that  $R$  is a binary relation on  $A$ .

✓ Ex :-  $A = \{0, 1, 2\}$  and  $B = \{3, 4, 5\}$

Let  $R = \{(1, 3) (2, 4) (2, 5)\}$



$R$  is a subset of  $A \times B$ . This is represented by the diagram (a) and this diagram is called the arrow diagram.

✓ Note :- If  $A$  has ' $m$ ' elements and  $B$  has ' $n$ ' elements, then the no. of relations from  $A$  to  $B$  is  $2^{mn}$ .

### Problems :-

1) Let  $A = \{1, 2, 3, 4, 6\}$  and  $R$  be the relation on  $A$  defined by  $(a, b) \in R$  if and only if  $a$  is a multiple of  $b$ . Write down  $R$  as a set of ordered pairs.

Soln :-  $R = \{(a, b) \mid a, b \in A \text{ and } a \text{ is a multiple of } b\}$   
 $R = \{(1, 1) (2, 1) (2, 2) (3, 1) (3, 3) (4, 1) (4, 2) (4, 4) (6, 1) (6, 2) (6, 3) (6, 6)\}$

2) Let  $A$  and  $B$  be finite sets with  $|B| = 3$ . If there are 4096 relations from  $A$  to  $B$ , what is  $|A| = ?$

Soln :- If  $|A| = m$ ,  $|B| = n$ , then there are  $2^{mn}$  relations from  $A$  to  $B$ .

Given  $|B| = n = 3$ . and  $2^{mn} = 4096$ .

$$\therefore 2^{3m} = 4096$$

$$\Rightarrow 3m \log_e 2 = \log_e 4096$$

$$\Rightarrow m = \frac{\log_e 4096}{3 \times \log_e 2} = 4$$

Thus  $|A| = 4$ .

3) Let  $A = \{1, 2, 3\}$  and  $B = \{2, 4, 5\}$ . Determine the following:

- i)  $|A \times B|$ .
- ii) No. of relations from  $A$  to  $B$ .
- iii) No. of binary relations on  $A$ .
- iv) No. of relations from  $A$  to  $B$  that contain  $(1, 2)$  &  $(1, 5)$ .
- v) No. of relations from  $A, B$  that contain exactly 5 ordered pairs.
- vi) No. of binary relations on  $A$  that contain atleast 7 ordered pairs.

Soln:- Given  $|A| = m = 3$ ,  $|B| = n = 3$ .

- i)  $|A \times B| = mn = 9$ .
- ii) No. of relations from  $A$  to  $B$  is  $2^{mn} = 2^9 = 512$ .
- iii) No. of binary relations on  $A$  is  $2^{mm} = 2^{m^2} = 2^9 = 512$ .
- iv) Let  $R_1 = \{(1, 2), (1, 5)\}$ . Every relation from  $A$  to  $B$  that contains the elements  $(1, 2)$  and  $(1, 5)$  is of the form  $R_1 \cup R_2$ , where  $R_2$  is a subset of  $\bar{R}_1$  in  $A \times B$ .  
 $\therefore$  No. of such relations = no. of subsets of  $\bar{R}_1$ .  
 $= 2^7$  ( $\because |\bar{R}_1| = |A \times B| - |R_1| = 9 - 2 = 7$ )  
 $= 128$

Thus there are 128 no. of relations from  $A$  to  $B$  that contain the elements  $(1, 2)$  and  $(1, 5)$ .

- v) Since  $A \times B$  contains 9 ordered pairs, the no. of relations from  $A$  to  $B$  that contain exactly 5 ordered pairs = no. of ways of choosing 5 ordered pairs from 9 ordered pairs.  
 This no. is  ${}^9C_5 = 126$ .

- vi) Similarly, the no. of binary relations on  $A$  that contain atleast 7 elements (ordered pairs) is  ${}^9C_7 + {}^9C_8 + {}^9C_9 = 46$ .

Matrix of a relation :- Consider the sets  $A = \{a_1, a_2, \dots, a_m\}$ .

$B = \{b_1, b_2, \dots, b_n\}$  of orders  $m$  and  $n$  resp. Then  $A \times B$  contains all ordered pairs of the form  $(a_i, b_j)$   $1 \leq i \leq m$ ,  $1 \leq j \leq n$  which are  $mn$  in number.

Let  $R$  be a relation from  $A$  to  $B$  so that  $R$  is a subset of  $A \times B$ . Let  $m_{ij} = (a_i, b_j)$  and

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases} \rightarrow \text{zero-one matrix}$$

The  $m \times n$  matrix formed by  $m_{ij}$  is called the relation matrix for  $R$  or the adjacency matrix or zero-one matrix for  $R$  and is denoted by  $M_R$  or  $M(R)$ .

Rows of  $M_R$  correspond to elements of  $A$  and columns correspond to elements of  $B$ .

When  $B = A$ , then  $M_R$  is an  $n \times n$  matrix whose elements

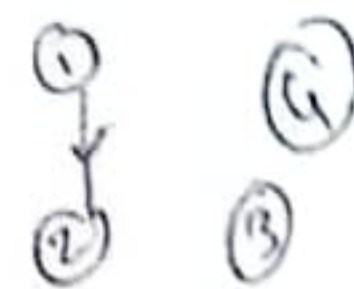
$$\text{are } m_{ij} = \begin{cases} 1 & \text{if } (a_i, a_j) \in R \\ 0 & \text{if } (a_i, a_j) \notin R \end{cases}$$

Ex:- 1)  $A = \{1, 2, 3, 4\}$   $B = \{4, 5\}$  and  $R = \{(1, 4) (2, 5)\}$

$$\therefore M_R = \begin{matrix} & & \begin{matrix} 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \end{matrix}$$

2)  $A = \{p, q, r\}$   $R = \{(p, p) (p, q) (r, r) (r, q)\}$  then.

$$M_R = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} p \\ q \\ r \end{matrix} & \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$



$\rightarrow$  is defined only on  $R: A \rightarrow A$ , never on  $R: A \rightarrow B$

Digraph of a relation :- Let  $R$  be a relation on a finite set  $A$ .

To get the digraph of  $R$ , we follow the following procedure :-

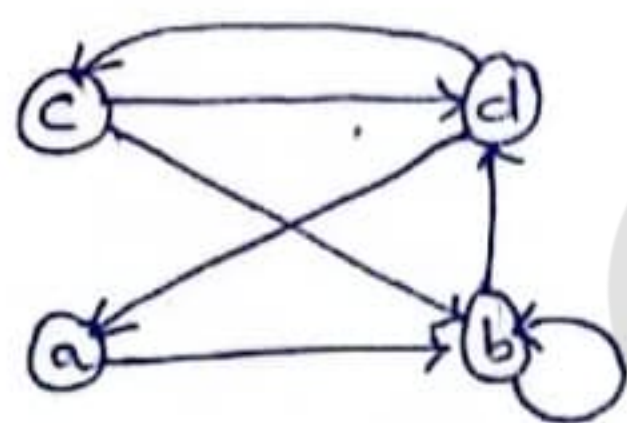
1. Draw a small circle or bullet for each of the elements of  $A$  and label it with the corresponding element of  $A$ . These circles are called vertices or nodes.

2. Draw an arrow, called an edge from a vertex  $x$  to a vertex  $y$  if and only if  $(x, y) \in R$ .

Note:-  $\rightarrow$  In a digraph, a vertex from which an edge leaves is called the origin or the source for that edge and a vertex where an edge ends is called the terminus for that edge.

- 2) A vertex which is neither a source nor a terminus of an edge is known as an isolated vertex.
- 3) An edge for which the source and the terminus are one and the same vertex is called a loop.
- 4) The no. of edges (arrows) terminating at a vertex is called the in-degree of that vertex and the no. of edges (arrows) leaving a vertex is called the out-degree of that vertex.

Ex:-  $\rightarrow A = \{a, b, c, d\}$   $R = \{(a, b) (b, b) (b, d) (c, b) (c, d) (d, a) (d, c)\}$



2)  $A = \{1, 2, 3, 4, 5\}$   $R = \{(1, 1) (1, 2) (1, 3) (1, 4) (3, 2)\}$



### Operations on Relations:

#### 1) Union and Intersection of Relations:

Given the relations  $R_1$  and  $R_2$  from a set  $A$  to a set  $B$ , the union of  $R_1$  and  $R_2$ , denoted by  $R_1 \cup R_2$ , is defined as a relation from  $A$  to  $B$  with the property that  $(a, b) \in R_1 \cup R_2$  iff  $(a, b) \in R_1$  or  $(a, b) \in R_2$ .

The intersection of  $R_1$  and  $R_2$ , denoted by  $R_1 \cap R_2$ , is defined as a relation from  $A$  to  $B$  with the property that  $(a, b) \in R_1 \cap R_2$

iff  $(a, b) \in R_1$  and  $(a, b) \in R_2$ .

### Complement of a Relation :

Given a relation  $R$  from a set  $A$  to a set  $B$ , the complement of  $R$ , denoted by  $\bar{R}$ , is defined as a relation from  $A$  to  $B$  with the property that  $(a,b) \in \bar{R}$  iff  $(a,b) \notin R$ .

### Converse of a Relation :-

Given a relation  $R$  from a set  $A$  to a set  $B$ , the converse of  $R$  denoted by  $R^c$ , is defined as a relation from  $B$  to  $A$  with the property that  $(a,b) \in R^c$  iff  $(b,a) \in R$ .

NOTE: 1) If  $M_R$  is the matrix of  $R$ , then  $(M_R)^T$ , the transpose of  $M_R$ , is the matrix of  $R^c$ .

2)  $(R^c)^c = R$ .

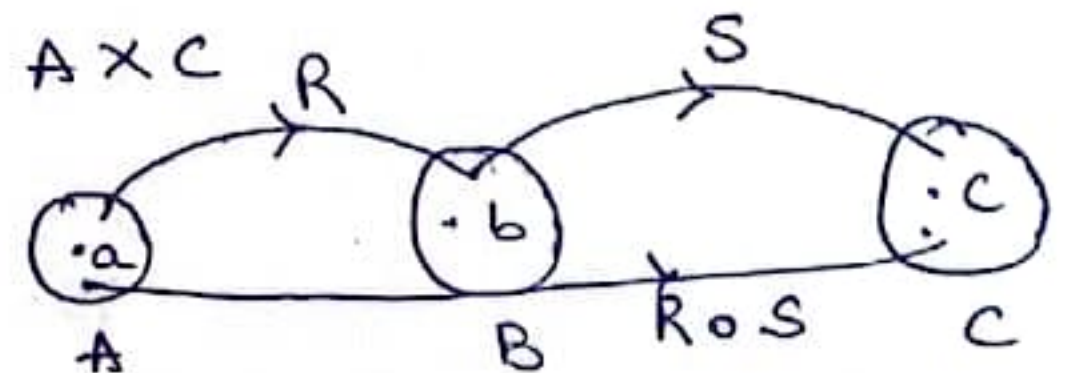
Composition of Relations :- consider a relation  $R$  from a set  $A$  to a set  $B$  and a relation  $S$  from the set  $B$  to the set  $C$ . Then the product or <sup>the</sup> composition of  $R$  and  $S$  is a relation from the set  $A$  to the set  $C$ , denoted by  $R \circ S$  and is defined as if  $a \in A$  and  $c \in C$  then  $(a,c) \in R \circ S$  iff there is some  $b$  in  $B$  such that  $(a,b) \in R$  and  $(b,c) \in S$ .

$\therefore R \circ S = \{(a,c) \mid a \in A, c \in C \text{ and } \exists b \in B \text{ with } (a,b) \in R \text{ and } (b,c) \in S\}$ .

NOTE:- 1)  $R \subseteq A \times B$ ,  $S \subseteq B \times C \Rightarrow R \circ S \subseteq A \times C$

2)  $R \circ S \neq S \circ R$ .

3) If  $R$  is a relation on  $A$ , then  $R \circ R$  is a relation on  $A$ , denoted by  $R^2$  and  $(R \circ R) \circ R$  is also a relation on  $A$ , denoted by  $R^3$ .



4) Let  $R$  be a relation from  $A$  to  $B$  and  $S$  be a relation from  $B$  to  $C$ , then the matrices of  $R$ ,  $S$  and  $R \circ S$  satisfy  $M(R) \cdot M(S) = M(R \circ S)$ .

5)  $M(R^2) = [M(R)]^2$  and  $M(R^n) = [M(R)]^n$ ,  $n \in \mathbb{Z}^+$ .

6) Let  $A, B, C, D$  be the sets and  $R, S, T$  be the relations from  $A$  to  $B$ ,  $B$  to  $C$  and  $C$  to  $D$  resp, then  $R \circ (S \circ T) = (R \circ S) \circ T$ .

Problems:-

1) Let  $A = \{1, 2, 3, 4\}$  and let  $R$  be the relation on  $A$  defined by  $x R y$  iff  $y = 2x$ .

i) write down  $R$  as a set of ordered pairs

ii) Draw the digraph of  $R$ .

iii) Determine the in-degrees and out-degrees of the vertices in the digraph. iv) write the matrix of  $R$ .

Soln:- i) for  $x, y \in A$ ,  $(x, y) \in R$  iff  $y = 2x$ .

$$\therefore R = \{(1, 2), (2, 4)\}.$$

ii) Digraph of  $R$  is as shown below:



| iii) | vertices | Indegree | out degree |
|------|----------|----------|------------|
|      | 1        | 0        | 1          |
|      | 2        | 1        | 1          |
|      | 3        | 0        | 0          |
|      | 4        | 1        | 0          |

$$iv) M(R) = M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

2) Let  $A = \{1, 2, 3, 4, 6\}$  and  $R$  be the relation on  $A$  defined by  $a R b$  iff  $a$  is a multiple of  $b$ .

i) Represent the relation  $R$  as a set of ordered pairs.

ii) Draw its digraph.

iii) Determine the indegrees and outdegrees of the vertices.

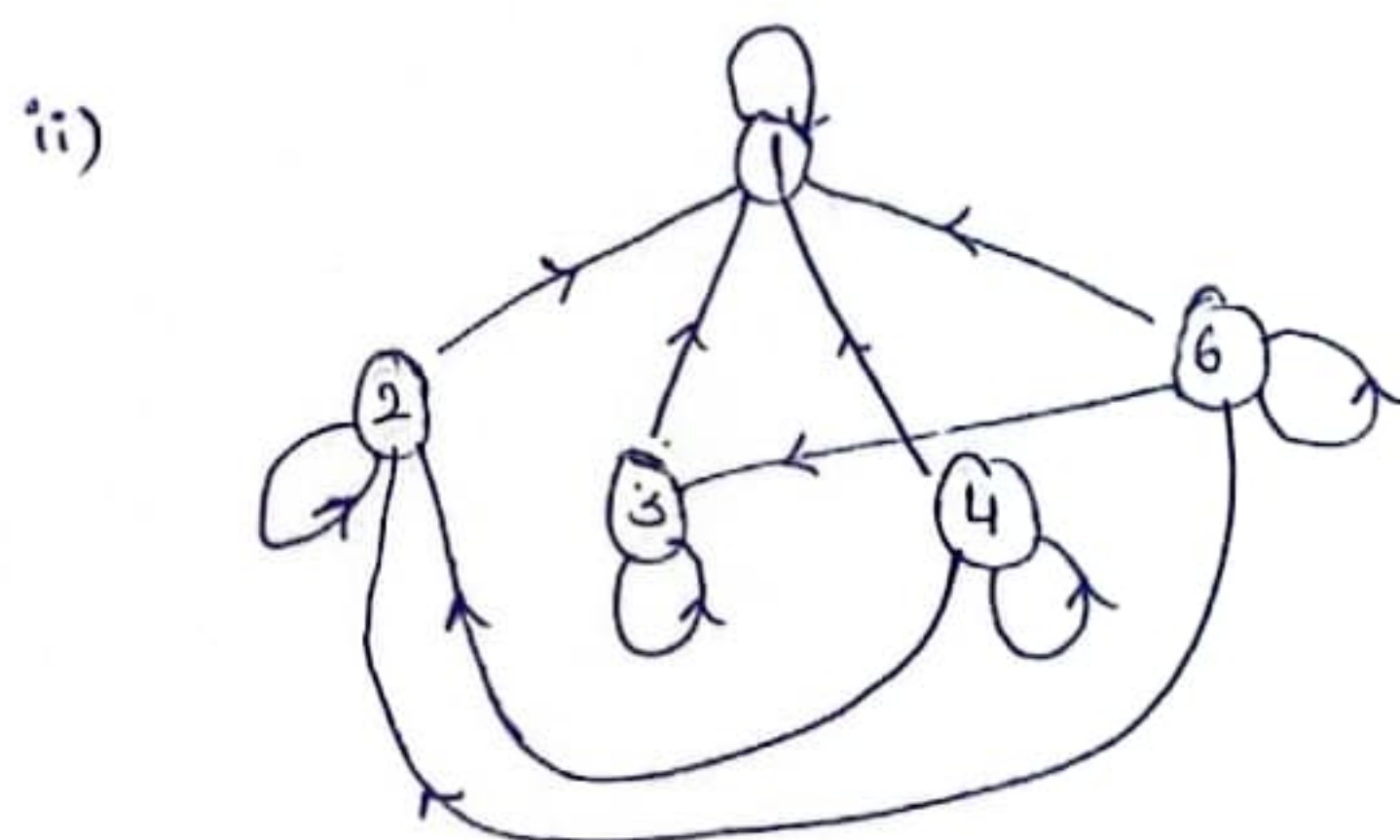
iv) write the matrix of  $R$ .

Soln:- Given  $(a, b) \in R$  iff  $a$  is a multiple of  $b$ .

$$i) R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 3), (4, 1), (4, 2), (4, 4), (6, 1), (6, 2), (6, 3), (6, 6)\}.$$

iv)  $M(R) =$

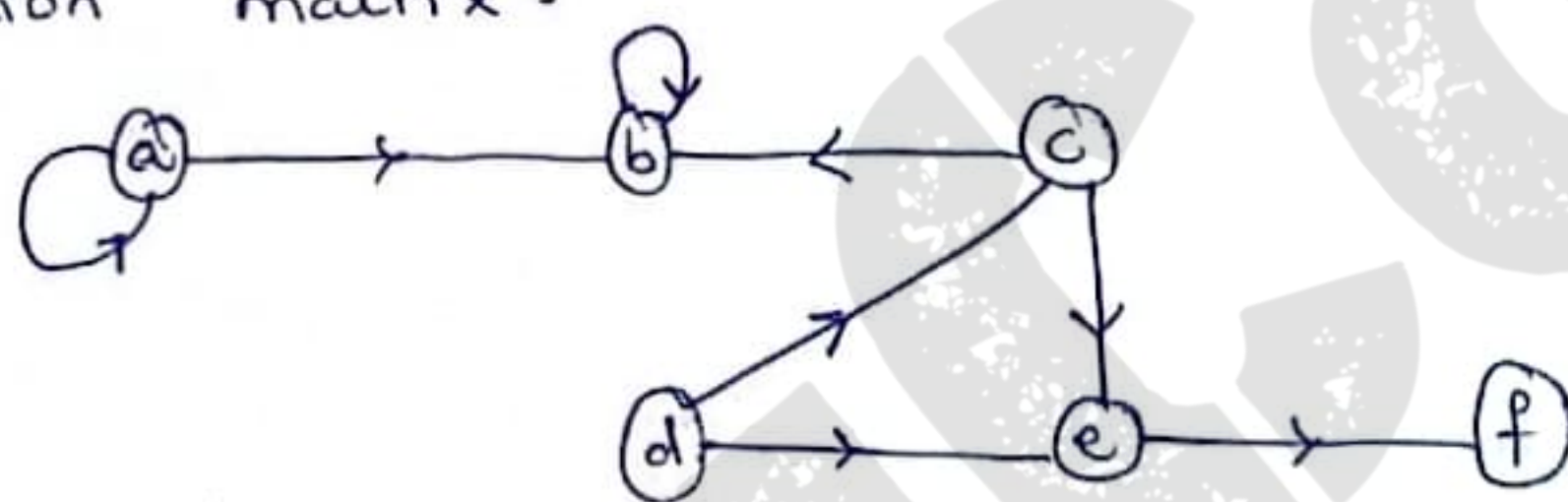
$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$



iii)

| vertices | Indegree | out degree |
|----------|----------|------------|
| 1        | 5        | 1          |
| 2        | 3        | 2          |
| 3        | 2        | 2          |
| 4        | 1        | 3          |
| 6        | 1        | 4          |

3) For  $A = \{a, b, c, d, e, f\}$  the digraph in the fig below represents a relation on  $A$ . Determine  $R$  and the associated relation matrix.



soln :-  $R = \{(a, a) (a, b) (b, b) (c, b) (c, e) (d, c) (d, e) (e, f)\}$ .

$$M(R) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

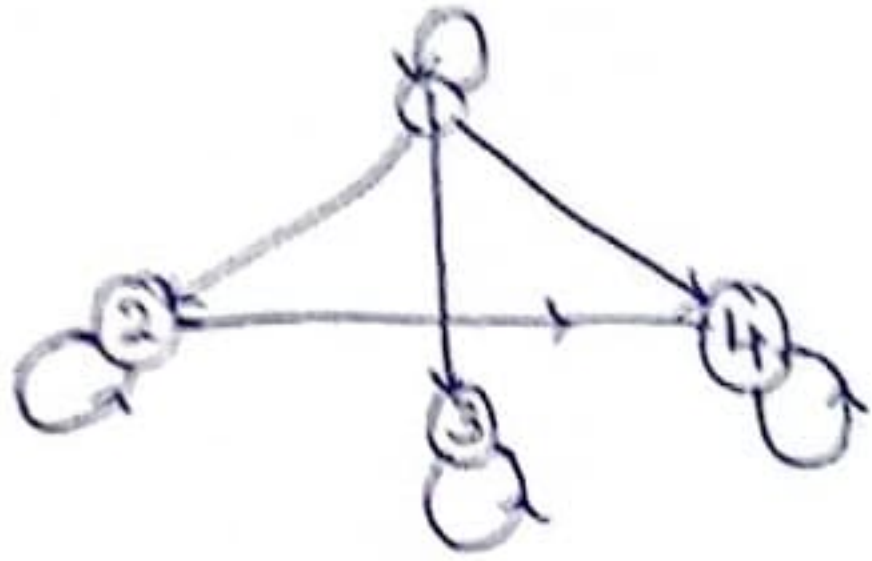
4) Let  $A = \{1, 2, 3, 4\}$  and  $R$  be a relation on  $A$  defined by  $(a, b) \in R$  iff  $a \mid b$  ( $a$  divides  $b$ ). write down  $R$  as a set of ordered pairs. write the matrix of this relation and draw the digraph of  $R$ . find indegree and outdegree of all the vertices. Also find  $R^2, R^3$  and matrices of  $R^2$  and  $R^3$  and digraphs.

P.T.O.

Soln:- Given  $A = \{1, 2, 3, 4\}$  and  $R$  is the relation on  $A$  defined by  $(a, b) \in R$  iff  $a|b$ .

$$R = \{(1, 1) (1, 2) (1, 3) (1, 4) (2, 2) (2, 4) (3, 3) (4, 4)\}$$

$$M(R) = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$



| vertices | Indegree | out degree |
|----------|----------|------------|
| 1        | 1        | 4          |
| 2        | 2        | 2          |
| 3        | 2        | 1          |
| 4        | 3        | 1          |

$$R^2 = R \circ R = \{(1, 1) (1, 2) (1, 3) (1, 4) (2, 2) (2, 4) (3, 3) (4, 4)\}$$

$$\{(1, 1) (1, 2) (1, 3) (1, 4) (2, 2) (2, 4) (3, 3) (4, 4)\}$$

$$R^2 = \{(1, 1) (1, 2) (1, 3) (1, 4) (2, 2) (2, 4) (3, 3) (4, 4)\}$$

$$\therefore \boxed{R^2 = R}$$

$$R^3 = R^2 \circ R = R \circ R = R^2 = R.$$

$$\therefore \boxed{R^3 = R}$$

Since  $R^2$  and  $R^3$  are same as  $R$ , matrices of  $R^2$  and  $R^3$  are same as matrix of  $R$  and digraphs of  $R^2$  and  $R^3$  are same as digraphs of  $R$ .  
Do 6th, then 5th.

5) Let  $A = \{1, 2, 3, 4\}$ ,  $B = \{w, x, y, z\}$  and  $C = \{5, 6, 7\}$ . Also let  $R_1$  be a relation from  $A$  to  $B$ ;  $R_2$  and  $R_3$  be relations from  $B$  to  $C$  defined by

$$R_1 = \{(1, x) (2, x) (3, y) (3, z)\}$$

$$R_2 = \{(w, 5) (x, 6)\}$$

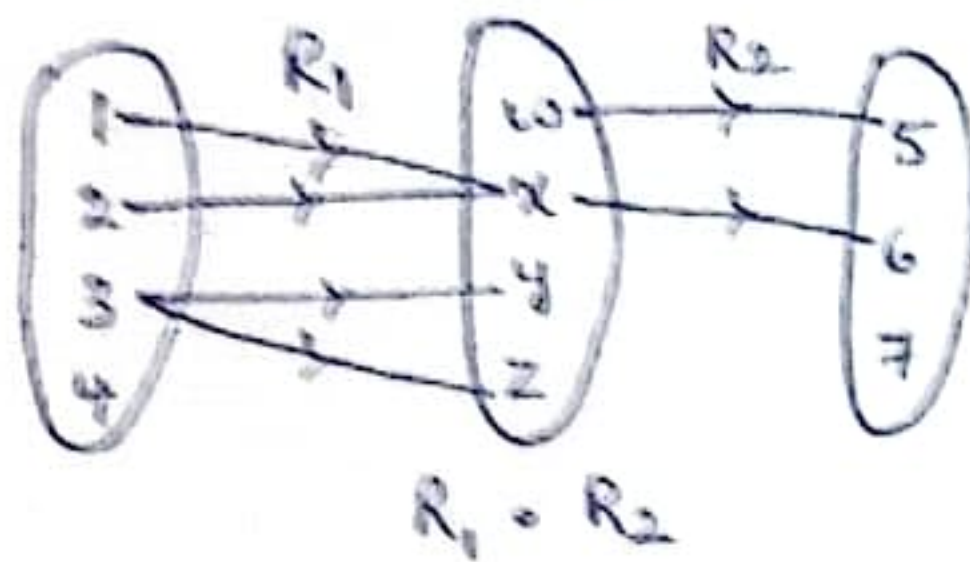
$$R_3 = \{(w, 5) (w, 6)\}.$$

i) Find  $R_1 \circ R_2$  and  $R_1 \circ R_3$ .

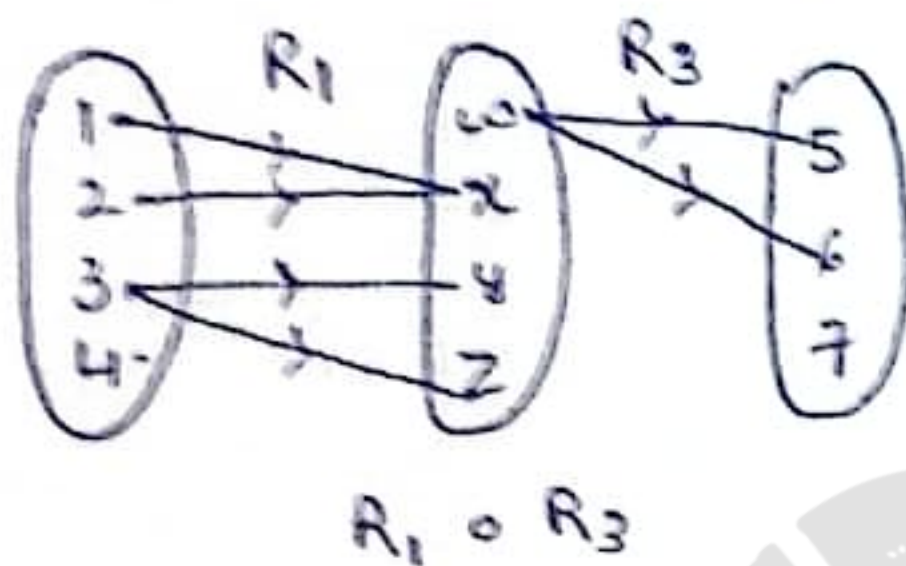
ii) Find  $M(R_1)$ ,  $M(R_2)$  and  $M(R_1 \circ R_2)$

iii) Verify that  $M(R_1 \circ R_2) = M(R_1) \cdot M(R_2)$

Soln:- i)  $(1, x) \in R_1$  and  $(x, 6) \in R_2 \Rightarrow (1, 6) \in R_1 \circ R_2$   
 $(2, x) \in R_1$  and  $(x, 6) \in R_2 \Rightarrow (2, 6) \in R_1 \circ R_2$ .  
 $\therefore R_1 \circ R_2 = \{(1, 6), (2, 6)\}$ . (see fig (i))



There is no element  $(a, b) \in R_1$  such that  $(b, c) \in R_3$   
 $\therefore R_1 \circ R_3 = \{\} = \phi$  (see fig (ii))



$$ii) M(R_1) = \begin{matrix} & \begin{matrix} w & x & y & z \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$M(R_2) = \begin{matrix} \begin{matrix} w & x & y & z \end{matrix} & \begin{matrix} 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

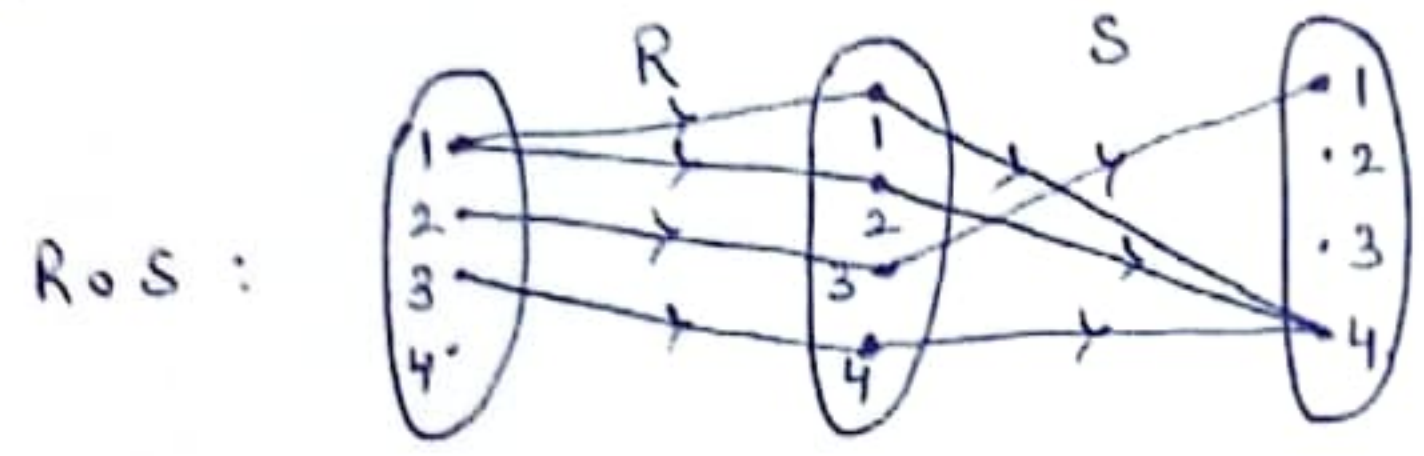
$$M(R_1 \circ R_2) = \begin{matrix} \begin{matrix} 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$iii) \text{ consider } M(R_1) \cdot M(R_2) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ = M(R_1 \circ R_2).$$

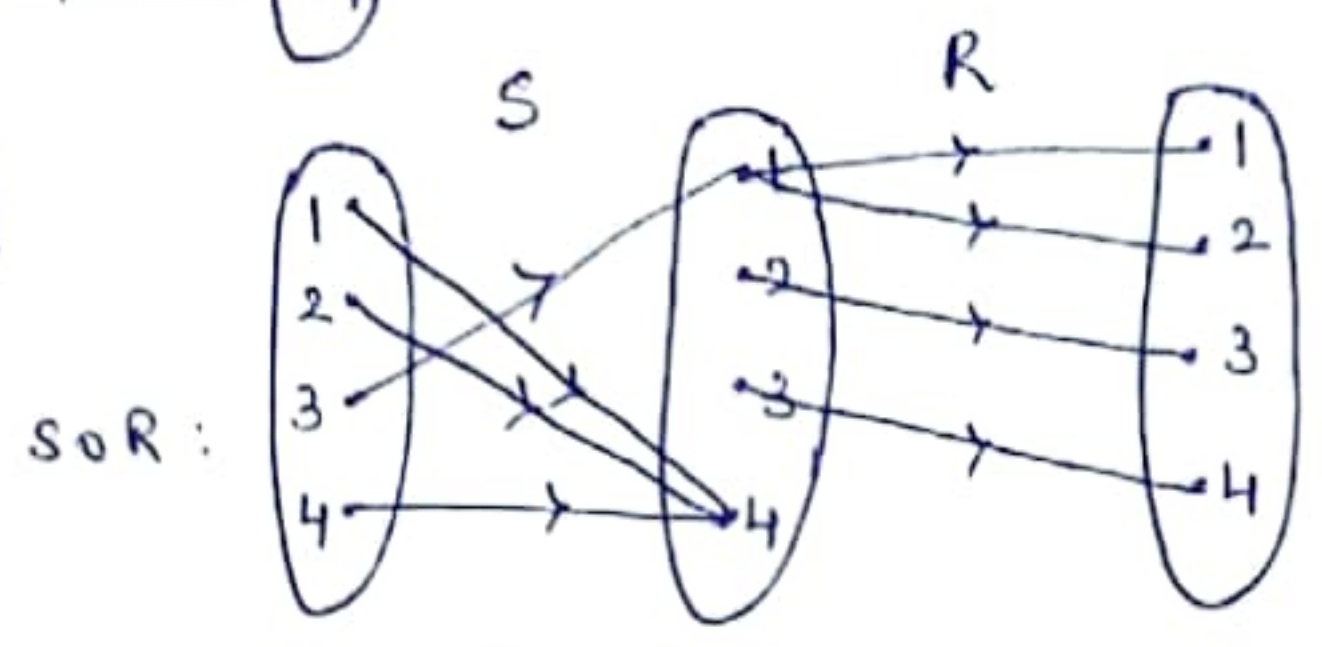
6) Let  $A = \{1, 2, 3, 4\}$  and  $R = \{(1, 1), (1, 2), (2, 3), (3, 4)\}$ ,  
 $S = \{(3, 1), (4, 4), (2, 4), (1, 4)\}$  be relations on  $A$ . Determine the  
relations  $R \circ S$ ,  $S \circ R$ ,  $R \circ (R \circ S)$ ,  $R \circ (S \circ R)$ ,  $S \circ (R \circ S)$ ,  $S \circ (S \circ R)$ ,  
 $R^2$  and  $S^2$ .

P.T.O.

Soln :-  $R \circ S = \{(1,4)(2,1)(3,4)\}$



$S \circ R = \{(3,1)(3,2)\}$



$R \circ (R \circ S) = \{(1,1)(1,2)(2,3)(3,4)\} \circ \{(1,4)(2,1)(3,4)\}$   
 $= \{(1,4)(1,1)(2,4)\}$

$R \circ (S \circ R) = \{(1,1)(1,2)(2,3)(3,4)\} \circ \{(3,1)(3,2)\}$   
 $= \{(2,1)(2,2)\}$

$S \circ (R \circ S) = \{(3,1)(4,4)(2,4)(1,4)\} \circ \{(1,4)(2,1)(3,4)\}$   
 $= \{(3,4)\}$

$S \circ (S \circ R) = \{(3,1)(4,4)(2,4)(1,4)\} \circ \{(3,1)(3,2)\}$   
 $= \phi = \{ \}$

$R^2 = R \circ R = \{(1,1)(1,2)(2,3)(3,4)\} \circ \{(1,1)(1,2)(2,3)(3,4)\}$   
 $= \{(1,1)(1,2)(1,3)(2,4)\}$

$S^2 = S \circ S = \{(3,1)(4,4)(2,4)(1,4)\} \circ \{(3,1)(4,4)(2,4)(1,4)\}$   
 $= \{(3,4)(4,4)(2,4)(1,4)\}$

7) If  $A = \{1,2,3,4\}$  and  $R$  is a relation on  $A$  defined by  
 $R = \{(1,2)(1,3)(2,4)(3,2)(3,3)(3,4)\}$ . find  $R^2$  and  $R^3$ . write down  
their digraphs. find  $M(R)$ ,  $M(R^2)$ ,  $M(R^3)$ . verify that  $M(R^2) = [M(R)]^2$   
and  $M(R^3) = [M(R)]^3$ .

Soln :-  $R^2 = R \circ R = \{(1,2)(1,3)(2,4)(3,2)(3,3)(3,4)\}$   
 $\circ$   
 $\{(1,2)(1,3)(2,4)(3,2)(3,3)(3,4)\}$

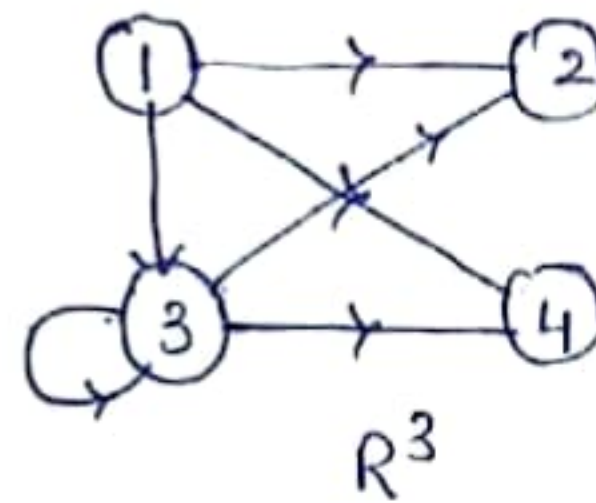
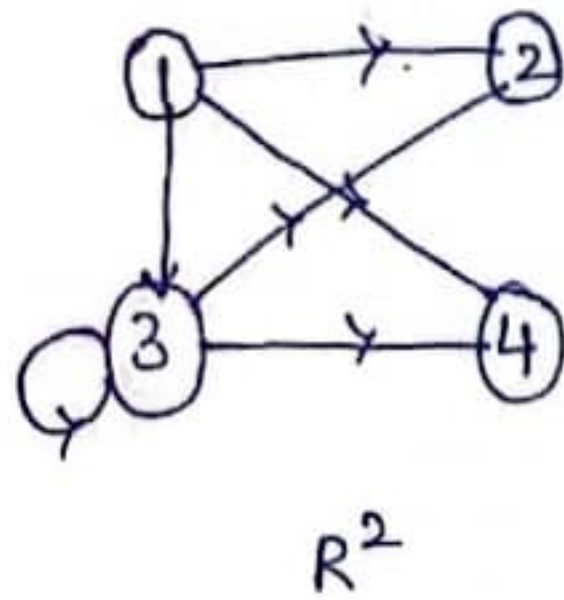
$$\therefore R^2 = \{(1,4) (1,2) (1,3) (3,4) (3,2) (3,3)\}.$$

$$R^3 = R^2 \circ R = \{(1,4) (1,2) (1,3) (3,4) (3,2) (3,3)\}$$

$$\{ (1,2) (1,3) (2,4) (3,2) (3,3) (3,4) \}$$

$$R^3 = \{(1,4) (1,2) (1,3), (3,4) (3,2) (3,3)\}.$$

The digraphs of  $R^2$  and  $R^3$  are as shown below:



$$M(R) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M(R^2) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

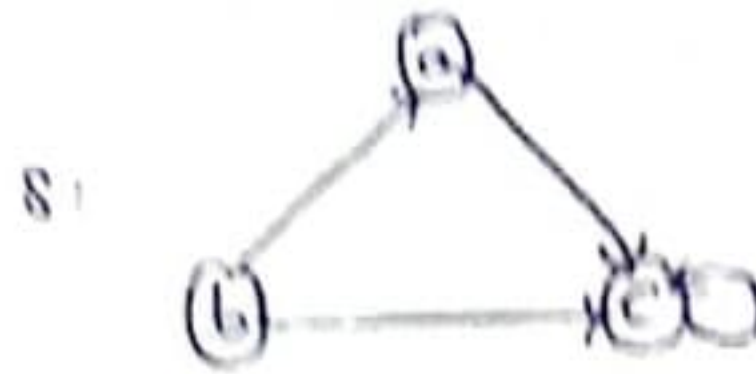
$$M(R^3) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

by matrix multiplication with the stipulation that  $1+1=1$ , we find

$$\begin{aligned} [M(R)]^2 &= M(R) \cdot M(R) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M(R^2). \end{aligned}$$

$$\begin{aligned} \text{and } [M(R)]^3 &= M(R) \cdot M(R^2) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M(R^3). \end{aligned}$$

8) The digraphs of two relations  $R$  and  $S$  on the set  $A = \{a, b, c\}$  are given below. Draw the digraphs of  $\bar{R}$ ,  $R \cup S$ ,  $R \cap S$  and  $R^c$ .



Soln :- From the digraphs,

$$R = \{(a, a), (a, b), (a, c), (b, c)\} \quad \& \quad S = \{(a, c), (b, a), (b, c), (c, c)\}$$

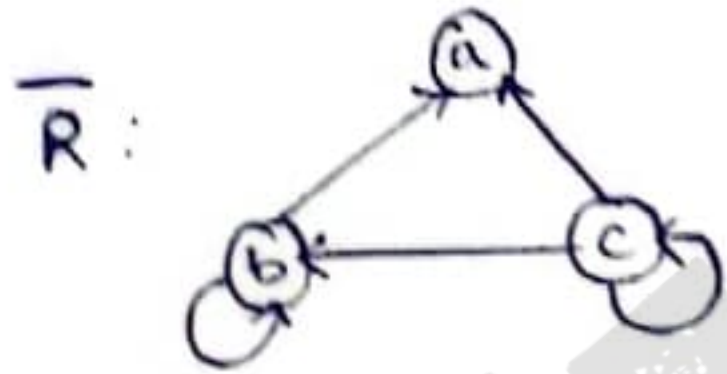
$$\therefore \bar{R} = \{(b, a), (b, b), (c, a), (c, b), (c, c)\}$$

$$R \cup S = \{(a, a), (a, b), (a, c), (b, a), (b, c), (c, c)\}$$

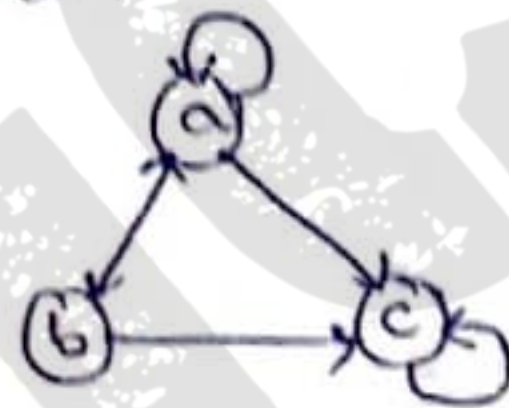
$$R \cap S = \{(a, c), (b, c)\}$$

$$R^c = \{(a, a), (b, a), (c, a), (c, b)\}$$

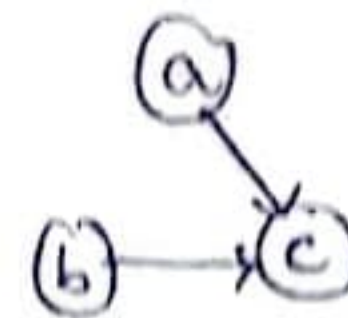
The digraphs of these are:



$R \cup S$ :



$R \cap S$ :



$R^c$ :



## Properties of Relations:

- 1) Reflexive relation: A relation  $R$  on a set  $A$  is said to be reflexive if  $(a, a) \in R \quad \forall \overset{\text{all } a}{a} \in A$ , i.e.  $aRa, \forall a \in A$ .  
 $R$  is not reflexive (i.e. non-reflexive) if  $\exists$  some  $a \in A$  such that  $(a, a) \notin R$ .

Ex:-  $\leq, =, \geq$  are reflexive relations on the set of all real no's. where  $<, >$  are not reflexive relations on the set of all real no's.

- 2) If  $A = \{1, 2, 3, 4\}$ , then  $R = \{(1, 1), (2, 2), (3, 3)\}$  is not non-reflexive b'coz  $4 \in A$  but  $(4, 4) \notin R$ .

- 2) Irreflexive relation:- A relation  $R$  on a set  $A$  is said to be irreflexive if  $(a, a) \notin R, \forall \overset{\text{all } a}{a} \in A$ . i.e. there is no element of  $A$  related to itself.

Ex:- ' $<$ ', ' $>$ ' are irreflexive on the set of all real no's.

Note:- 1) A non reflexive relation need not be irreflexive.

2) A relation can be neither reflexive nor irreflexive.

- 3) Symmetric Relation:- A relation  $R$  on a set  $A$  is said to be symmetric whenever  $(a, b) \in R, (b, a) \in R \quad \forall a, b \in A$ .

- 4) Asymmetric relation:- A relation  $R$  on a set  $A$  is said to be asymmetric whenever  $(a, b) \in R, (b, a) \notin R \quad \forall a, b \in A$ .

- 5) Antisymmetric relation:- A relation  $R$  on a set  $A$  is said to be anti-symmetric if whenever  $(a, b) \in R$  and  $(b, a) \in R$  then  $a = b$ . i.e.  $(a, b) \in R$ , and  $a \neq b$  then  $(b, a) \notin R$ .

Ex:- ' $\leq$ ' is antisymmetric on the set of all real numbers, b'coz if  $a \leq b$  and  $b \leq a$ , then  $a = b$ .

Note: 1) Asymmetric and antisymmetric relations are not same.

2) A rel<sup>n</sup> can be both symmetric and antisymmetric. It can be neither symmetric nor antisymmetric.

6) Transitive relation :- A relation  $R$  on a set  $A$  is said to be transitive if whenever  $(a, b) \in R$  and  $(b, c) \in R$  then  $(a, c) \in R \quad \forall a, b, c \in A$ .

Ex:- ' $\leq$ ', ' $>$ ' are transitive on the set of all real no's, b'coz if  $a \leq b$  and  $b \leq c$  then  $a \leq c$  and if  $a > b$  and  $b > c$  then  $a > c$ .  $\forall a, b, c \in \mathbb{R}$  (real no's)

Note:-  $R$  is not transitive if there exist  $a, b, c \in A$  such that  $(a, b) \in R$  and  $(b, c) \in R$  but  $(a, c) \notin R$ .

— X —

Equivalence relation :- A relation  $R$  on a set  $A$  is said to be an equivalence relation on  $A$  if  $R$  is reflexive, symmetric and transitive on  $A$ .

Ex:- ' $=$ ' is an equivalence relation on the set of all real no's.

Equivalence classes :- Let  $R$  be an equivalence relation on a set  $A$  and  $a \in A$ . Then the set of all those elements of  $A$  which are related to  $a$  by  $R$  is called the equivalence class of  $a$  with respect to  $R$ . This equivalence class is denoted by  $R(a)$  or  $[a]$  or  $\bar{a}$ . Thus  $\bar{a} = [a] = R(a) = \{x \in A \mid (x, a) \in R\}$ .

Ex:-  $R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$   $[1] = \{1, 2\}$   $[2] = \{1, 2\}$   $[3] = \{3\}$   $\Downarrow xRa$

Partition of a set :- Let  $A$  be a non empty set. Suppose there exist non empty subsets  $A_1, A_2, A_3, \dots, A_K$  of  $A$  such that the follow-

ing two conditions hold :

- i)  $A$  is union of  $A_1, A_2, \dots, A_K$  i.e.  $A = A_1 \cup A_2 \cup \dots \cup A_K$ .
- ii) Any 2 of the subsets  $A_1, A_2, \dots, A_K$  are disjoint i.e.  $A_i \cap A_j = \phi$  for  $i \neq j$ .

Then  $P = \{A_1, A_2, \dots, A_K\}$  is called a Partition of  $A$ . Also  $A_1, A_2, \dots, A_K$  are called the blocks or cells of the Partition. A Partition of a set with 6 blocks (cells) is as shown below:



Problem:

Let  $A = \{1, 2, 3\}$ . Determine the nature of the following relations on  $A$ .

i)  $R_1 = \{(1, 2) (2, 1) (1, 3) (3, 1)\}$ .

- Irreflexive
- symmetric
- non-transitive.

ii)  $R_2 = \{(1, 1) (2, 2) (3, 3) (2, 3)\}$ .

- Reflexive
- Transitive
- Asymmetric (not symmetric)

iii)  $R_3 = \{(1, 1) (2, 2) (3, 3)\}$

- Reflexive
  - symmetric
  - transitive
- } Equivalence relation

iv)  $R_4 = \{(1, 1) (2, 2) (3, 3) (2, 3) (3, 2)\}$

- Reflexive
  - symmetric
  - Transitive
- } Equivalence Relation

v)  $R_5 = \{(1, 1) (2, 3) (3, 3)\}$

- Non-reflexive or Irreflexive
- not symmetric i.e. Asymmetric.
- Transitive

vi)  $R_6 = \{(2, 3) (3, 4) (2, 4)\}$ .

- Transitive
- Irreflexive.
- Not symmetric

vii)  $R_7 = \{(1, 3) (3, 2)\}$

- Irreflexive
- non-transitive
- not symmetric

2) If  $A = \{1, 2, 3, 4\}$  and  $R$  be a relation on  $A$ .  
 Give an example of a relation for each of the following:

- i) reflexive and symmetric, but not transitive
- ii) reflexive and transitive, but not symmetric
- iii) symmetric and transitive, but not reflexive.

Soln:- i)  $\{(1,1) (2,2) (3,3) (4,4) (1,2) (2,1) (2,3) (3,2)\}$ .

ii)  $\{(1,1) (2,2) (3,3) (4,4) (1,2)\}$

iii)  $\{(1,1) (2,2) (1,2) (2,1)\}$ .

3) Let  $A = \{1, 2, 3, 4\}$  and  $R = \{(1,1) (1,2) (2,1) (2,2) (3,4) (4,3) (3,3) (4,4)\}$  be a relation on  $A$ . Verify that  $R$  is an equivalence relation.

Soln:- To verify  $R$  is an equivalence relation, we have to show that  $R$  is reflexive, symmetric and transitive.

i)  $(1,1) (2,2) (3,3) (4,4)$  belong to  $R$ .  $\therefore (a,a) \in R \forall a \in A$ .

$\therefore R$  is reflexive.

ii)  $(1,2), (2,1) \in R$  and  $(3,4) (4,3) \in R$

$\therefore$  if  $(a,b) \in R$ , then  $(b,a) \in R$  for  $a, b \in A$ .

$\therefore R$  is symmetric.

iii)  $(1,2) (2,1) (1,1) \in R$ ,  $(2,1) (1,2) (2,2) \in R$ ,

$(4,3) (3,4) (4,4) \in R$ .

$\therefore$  if  $(a,b) \in R$  and  $(b,c) \in R$ , then  $(a,c) \in R \forall a, b, c \in A$

$\therefore R$  is transitive.

Thus  $R$  is an equivalence relation.

→ Do this problem after 7<sup>th</sup>, combining with 8<sup>th</sup>.

4) Let  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ . On this set define the relation  $R$  by  $(x,y) \in R$  iff  $x-y$  is a multiple of 5.

Verify that  $R$  is an equivalence relation.

Soln:- i) for any  $x \in A$ , we have  $x-x=0$  is a multiple of 5

$(\because 0=5 \times 0) \therefore (x,x) \in R$ .

$\therefore R$  is reflexive.

ii) for any  $x, y \in A$ ,

if  $(x, y) \in R$  then  $x - y = 5k$  for some integer  $k$ .

$\Rightarrow y - x = 5(-k)$  so that  $(y, x) \in R$ .

$\therefore R$  is symmetric.

iii) for any  $x, y, z \in A$ , if  $(x, y) \in R$  and  $(y, z) \in R$ , then  
 $x - y = 5k_1$  and  $y - z = 5k_2$  for some integers  $k_1$  and  $k_2$ .

$$\begin{aligned} \therefore x - z &= (x - y) - (z - y) \\ &= (x - y) + (y - z) \\ &= 5k_1 + 5k_2 = 5(k_1 + k_2) \end{aligned}$$

$\therefore (x, z) \in R \Rightarrow R$  is Transitive.

Thus  $R$  is an equivalence relation.

5) Ex  $A = A_1 \cup A_2 \cup A_3$ , where  $A_1 = \{1, 2\}$ ,  $A_2 = \{2, 3, 4\}$  and  $A_3 = \{5\}$

Define the relation  $R$  on  $A$  by  $xRy$  iff  $x$  and  $y$  are in the same set  $A_i$ ,  $i = 1, 2, 3$ . Is  $R$  an equivalence relation?

Soln:-  $A = A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4, 5\}$

$$R = \{(1, 1) (1, 2) (2, 1) (2, 2) (2, 3) (2, 4) (3, 2) (3, 3) (3, 4) (4, 2) (4, 3) (4, 4) (5, 5)\}$$

i)  $R$  is reflexive  $\because (a, a) \in R \forall a \in A$ .

ii)  $R$  is symmetric  $\because$  whenever  $(a, b) \in R$ ,  $(b, a) \in R \forall a, b \in A$ .

iii)  $R$  is not transitive  $\because (1, 2) \in R$ ,  $(2, 3) \in R$  but  $(1, 3) \notin R$ .

$\therefore R$  is not an equivalence relation.

6) for a fixed integer  $n > 1$ , Prove that the relation "Congruent modulo  $n$ " is an equivalence relation on the set of all integers  $\mathbb{Z}$ .

Soln:- For any  $a, b \in \mathbb{Z}$ , "a is congruent to b modulo  $n$ "  $[a \equiv b \pmod{n}]$

if  $a - b$  is a multiple of  $n$ . i.e.  $a - b = kn$  for some  $k \in \mathbb{Z}$ .

Let  $R$  be a relation on  $\mathbb{Z}$  defined by  $a R b$  iff  $a \equiv b \pmod{n}$

i) we have  $a - a = 0 = 0 \times n \Rightarrow a \equiv a \pmod{n}$ .

$\therefore a R a$ .

$\Rightarrow R$  is reflexive.

ii) Since  $a R b$ ,  $a \equiv b \pmod{n}$

$$\therefore a - b = kn$$

$$\Rightarrow b - a = (-k)n$$

$$\Rightarrow b \equiv a \pmod{n} \quad (\because k \in \mathbb{Z}, -k \in \mathbb{Z})$$

$$\therefore b R a$$

Thus whenever  $a R b$ ,  $b R a \quad \forall a, b \in \mathbb{Z}$ .

$\therefore R$  is symmetric.

iii) Let  $a R b$  and  $b R c$ .

$$\text{ii} \quad a \equiv b \pmod{n} \quad \text{and} \quad b \equiv c \pmod{n}$$

$$\text{ii} \quad a - b = k_1 n \rightarrow \textcircled{1} \quad \text{and} \quad b - c = k_2 n \rightarrow \textcircled{2}, \quad k_1, k_2 \in \mathbb{Z}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow a - c = k_1 n + k_2 n$$

$$a - c = (k_1 + k_2)n$$

$$\Rightarrow a \equiv c \pmod{n} \quad (\because k_1 + k_2 \in \mathbb{Z})$$

$$\therefore a R c$$

Thus  $a R b$  and  $b R c \Rightarrow a R c, \quad \forall a, b, c \in \mathbb{Z}$ .

$\therefore R$  is transitive.

Thus  $R$  is an equivalence relation.

$\Rightarrow$  For the equivalence relation,  $R = \{(1,1)(1,2)(2,1)(2,2)(3,4)(4,3)(3,3)(4,4)\}$  defined on the set  $A = \{1,2,3,4\}$ . Determine the Partition induced.

Soln:- The equivalence classes of the elements of  $A$  w.r.t  $R$  are

$$[1] = \{1,2\} \quad [2] = \{1,2\} \quad [3] = \{3,4\} \quad [4] = \{3,4\}$$

of these, only  $[1]$  and  $[3]$  are distinct.

$\therefore$  Partition  $P = \{[1], [3]\} = \{\{1,2\}, \{3,4\}\}$  is the Partition of the given  $A$  induced by  $R$ .

$$[\text{observe that } A = [1] \cup [3] = \{1,2\} \cup \{3,4\} = \{1,2,3,4\}]$$

P.T.O.

8) Find the Partition of A induced by R, given

$$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Soln:-  $R = \{(x, y) \in R \mid x - y \text{ is a multiple of } 5\}.$

$$\therefore R = \{(1,1) (2,2) \dots (12,12) (6,1) (7,2) (8,3) (9,4) (10,5) (11,6) (12,7) (11,1) (12,2) (1,6) (2,7) (3,8) (4,9) (5,10) (6,11) (7,12) (1,11) (2,12)\} \text{ (or) [see (*)]}$$

$\therefore$  Equivalence classes are

$$[1] = \{1, 6, 11\} \quad [2] = \{2, 7, 12\} \quad [3] = \{3, 8\}$$

$$[4] = \{4, 9\} \quad [5] = \{5, 10\}.$$

All these classes are distinct.

$\therefore P = \{[1], [2], [3], [4], [5]\}$  is the Partition of A induced

by R, and  $A = \{1, 6, 11\} \cup \{2, 7, 12\} \cup \{3, 8\} \cup \{4, 9\} \cup \{5, 10\}.$

9) Let  $A = \{1, 2, 3, 4, 5, 6, 7\}$  and R be the equivalence relation on A that induces the Partition  $A = \{1, 2\} \cup \{3\} \cup \{4, 5, 7\} \cup \{6\}.$  Find R.

Soln:- Given Partition has 4 blocks:  $\{1, 2\}, \{3\}, \{4, 5, 7\}, \{6\}.$

Let R be the equivalence relation <sup>Inducing</sup> ~~including~~ this Partition.

Since 1, 2 are in the same block, we have  $1R1, 1R2, 2R1, 2R2.$

Since 3 belongs to block  $[3]$ ,  $3R3.$

Since 4, 5, 7 belongs to same block,

$$4R4, 4R5, 4R7, 5R4, 5R5, 5R7, 7R4, 7R5, 7R7.$$

Since 6 belongs to  $[6]$ ,  $6R6.$

$$\therefore R = \{(1,1) (1,2) (2,1) (2,2) (3,3) (4,4) (4,5) (4,7) (5,4) (5,5) (5,7) (7,4) (7,5) (7,7) (6,6)\}.$$

(\*) Prob 8) by defn  $[a] = \{x \in A \mid xRa\}.$

$$\therefore [1] = \{x \in A \mid xR1\} = \{x \in A \mid x-1 \text{ is a multiple of } 5\}$$

$$= \{1, 6, 11\}.$$

10) Let  $A = \{1, 2, 3, 4, 5\}$ . Define a relation  $R$  on  $A \times A$  by

$(x_1, y_1) R (x_2, y_2)$  if and only if  $x_1 + y_1 = x_2 + y_2$ .

- i) verify that  $R$  is an equivalence relation on  $A \times A$ .
- ii) Determine the equivalence classes  $[(1, 3)]$ ,  $[(2, 4)]$  and  $[(1, 1)]$ .
- iii) Determine the partition of  $A \times A$  induced by  $R$ .

Soln:-

i) a) for any  $(x, y) \in A \times A$ , we have

$$x + y = x + y.$$

$$\Rightarrow (x, y) R (x, y). \quad \Rightarrow R \text{ is reflexive.}$$

b) for any  $(x_1, y_1), (x_2, y_2) \in A \times A$ ,

Suppose  $(x_1, y_1) R (x_2, y_2)$  then  $x_1 + y_1 = x_2 + y_2$ .

$$\Rightarrow x_2 + y_2 = x_1 + y_1.$$

$$\Rightarrow (x_2, y_2) R (x_1, y_1)$$

$\therefore R$  is symmetric.

c) for any  $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in A \times A$ ,

suppose  $(x_1, y_1) R (x_2, y_2)$  and  $(x_2, y_2) R (x_3, y_3)$

then  $x_1 + y_1 = x_2 + y_2$  and  $x_2 + y_2 = x_3 + y_3$

$$\Rightarrow x_1 + y_1 = x_3 + y_3.$$

$$\Rightarrow (x_1, y_1) R (x_3, y_3).$$

$\therefore R$  is transitive.

Thus  $R$  is an equivalence relation.

$$\begin{aligned} \text{ii) we have } [(1, 3)] &= \{(x, y) \in A \times A \mid (x, y) R (1, 3)\} \\ &= \{(x, y) \in A \times A \mid x + y = 1 + 3 = 4\} \\ &= \{(1, 3), (2, 2), (3, 1)\} \quad [\because A = \{1, 2, 3, 4, 5\}] \end{aligned}$$

$$\text{iii) } [(2, 4)] = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

$$[(1, 1)] = \{(1, 1)\}$$

iii) To determine the partition induced by  $R$ , we have to first find the equivalence classes of all elements  $(x, y)$  of  $A \times A$  w.r.t  $R$ .

P.T.O.

we have  $[(1,1)] = \{(1,1)\}$ .

$$[(1,2)] = \{(1,2) (2,1)\} = [(2,1)]$$

$$[(1,3)] = \{(1,3) (3,1), (2,2)\} = [(3,1)] = [(2,2)]$$

$$[(1,4)] = \{(1,4) (4,1) (2,3) (3,2)\} = [(4,1)] = [(2,3)] = [(3,2)]$$

$$[(2,4)] = \{(2,4) (4,2) (3,3)\}$$

$$[(1,5)] = \{(1,5) (5,1) (3,3) (2,4) (4,2)\} = [(5,1)] = [(3,3)] = [(2,4)] = [(4,2)]$$

$$[(2,5)] = \{(2,5) (5,2) (3,4) (4,3)\} = [(5,2)] = [(3,4)] = [(4,3)]$$

$$[(3,5)] = \{(3,5) (5,3) (4,4)\} = [(4,4)] = [(5,3)]$$

$$[(4,5)] = \{(4,5) (5,4)\} = [(5,4)]$$

$$[(5,5)] = \{(5,5)\}$$

Thus  $[(1,1)]$ ,  $[(1,2)]$ ,  $[(1,3)]$ ,  $[(1,4)]$ ,  $[(1,5)]$ ,  $[(2,5)]$ ,  $[(3,5)]$ ,  $[(4,5)]$ ,  $[(5,5)]$  are the only distinct equivalence classes of

$A \times A$  on  $R$ .

$\therefore$  Partition of  $A \times A$  induced by  $R$  is represented by

$$P = \{[(1,1)], [(1,2)], [(1,3)], [(1,4)], [(1,5)], [(2,5)], [(3,5)], [(4,5)], [(5,5)]\}$$

$$(A \times A = [(1,1)] \cup [(1,2)] \cup [(1,3)] \cup [(1,4)] \cup [(1,5)] \cup [(2,5)] \cup [(3,5)] \cup [(4,5)] \cup [(5,5)])$$

11) Find the number of equivalence relations that can be defined on a finite set  $A$  with  $|A| = 6$ .  $\rightarrow$  DO in first date

Soln:-

Note:- The no. of possible ways to assign 'm' distinct objects into 'n' identical places with empty places allowed is given by the formula  $p(m) = \sum_{i=1}^n S(m, i)$  for  $m \geq n$ , where  $S(m, n)$  is

the Stirling number of 2nd kind given by

$$S(m, n) = \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m \text{ for } m \geq n.$$

This no. represents the no. of ways of arranging 'm' objects into 'n' distinct containers with no container left empty.

Soln: Since  $|A| = 6$ , the partition of  $A$  can have atmost 6 cells.

Treating the elements of  $A$  as objects (i.e.  $m = 6$ ) and cells as containers (i.e.  $n = 6$ ), the no. of partitions having  $k$  cells is  $S(6, k)$ . Since  $k$  varies from 1 to 6, the total no.

of different partitions of  $A$  is

$$P(6) = \sum_{i=1}^6 S(6, i) = S(6, 1) + S(6, 2) + S(6, 3) + S(6, 4) + S(6, 5) + S(6, 6)$$

$$\therefore S(m, n) = \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m$$

$$\begin{aligned} S(6, 1) &= \frac{1}{1!} \sum_{k=0}^1 (-1)^k \binom{1}{1-k} (1-k)^6 \\ &= 1 \cdot 1C_1 \cdot 1^6 + (-1) \cdot 1C_0 \cdot (0)^6 \\ &= 1 + 0 = 1 \end{aligned}$$

$$\begin{aligned} S(6, 2) &= \frac{1}{2!} \sum_{k=0}^2 (-1)^k \binom{2}{2-k} (2-k)^6 \\ &= \frac{1}{2} [1 \cdot 2^6 - 2C_1 \cdot 1^6 + 2C_0 \cdot (0)^6] \\ &= \frac{1}{2} [2^6 - 2] = 31 \end{aligned}$$

$$\begin{aligned} S(6, 3) &= \frac{1}{3!} \sum_{k=0}^3 (-1)^k \binom{3}{3-k} (3-k)^6 \\ &= \frac{1}{6} [3^6 - 3 \times 2^6 + 3 \times 1^6] = 90 \end{aligned}$$

$$\begin{aligned} S(6, 4) &= \frac{1}{4!} \sum_{k=0}^4 (-1)^k \binom{4}{4-k} (4-k)^6 \\ &= \frac{1}{24} [4^6 - 4 \times 3^6 + 6 \times 2^6 - 4 \times 1^6] = 65 \end{aligned}$$

$$\begin{aligned} S(6, 5) &= \frac{1}{5!} \sum_{k=0}^5 (-1)^k \binom{5}{5-k} (5-k)^6 \\ &= \frac{1}{120} [5^6 - 5 \times 4^6 + 10 \times 3^6 - 10 \times 2^6 + 5 \times 1^6] = 15 \end{aligned}$$

$$\begin{aligned} S(6, 6) &= \frac{1}{6!} \sum_{k=0}^6 (-1)^k \binom{6}{6-k} (6-k)^6 \\ &= \frac{1}{720} [6^6 - 6 \times 5^6 + 15 \times 4^6 - 20 \times 3^6 + 15 \times 2^6 - 6 \times 1^6 + 0] = 1 \end{aligned}$$

$\therefore$  No. of Partitions of  $A$  is

$$P(6) = 1 + 31 + 90 + 65 + 15 + 1 = 203 \text{ i.e. } 203 \text{ equivalence relations can be defined on } A.$$

Partial Order :- A relation  $R$  on a set  $A$  is said to be a partial order on  $A$ , if  $R$  is reflexive, antisymmetric and transitive on  $A$ . A set  $A$  with a partial order defined on it is called a Partially order set (or) a Poset and is denoted by the pair  $(A, R)$ .

Ex:- The relation ' $\leq$ ' on the set of all integers is a Partial order.  $\therefore (\mathbb{Z}, \leq)$  is a Poset. The relation ' $\geq$ ' on the set of all integers is a partial order.  $\therefore (\mathbb{Z}, \geq)$  is a Poset.

Total order : Let  $R$  be a partial order on a set  $A$ .

Then  $R$  is called a total order on  $A$ , if for all  $x, y \in A$ , either  $xRy$  or  $yRx$ . In this case, the Poset  $(A, R)$  is called a totally ordered set.

Ex:- The Partial order relation ' $\leq$ ' is a total order on the set of all real numbers  $\mathbb{R}$ , because for any  $x, y \in \mathbb{R}$ , we have  $x \leq y$  or  $y \leq x$ .  $\therefore (\mathbb{R}, \leq)$  is a totally ordered set.

Note:- Every total order is a partial order but every partial order need not be a total order.

Hasse Diagrams :-

For a Partial order relation on a finite set, we can draw digraph of a Partial order.

1. Since a Partial order is reflexive, at every vertex of digraph of Partial order, there will be a loop. While drawing the digraph of partial order, we do not show the loops explicitly. They will be automatically understood by convention.
2. In the digraph of Partial order, if there is an edge from vertex  $a$  to  $b$  and an edge from vertex  $b$  to  $c$ , then there will be an edge from  $a$  to  $c$  (b'coz of transitivity). But we do not exhibit an edge from  $a$  to  $c$  explicitly. It will be automatically understood by convention.

3. To simplify the format of the digraph of a partial order, we represent the vertices by dots and draw the digraph in such a way that all edges point upward. With this convention we need not put arrows in the edges.

The digraph of a partial order drawn by adopting the conventions indicated above is called a Poset diagram or the Hasse diagram for the Partial order.

Problems:-

1) Let  $A = \{1, 2, 3, 4\}$  and  $R = \{(1,1)(1,2)(2,2)(2,4)(1,3)(3,3)(3,4), (1,4)(4,4)\}$ . Verify that  $R$  is a Partial order on  $A$ .

Also write down the Hasse diagram for  $R$ .

Soln:- i) <sup>we see that</sup>  $(a,a) \in R \quad \forall a \in A$ . Hence  $R$  is reflexive on  $A$ .

ii) if  $(a,b) \in R$  and  $a \neq b$ , then <sup>we see that</sup>  $(b,a) \notin R, \quad \forall a, b \in A$ .

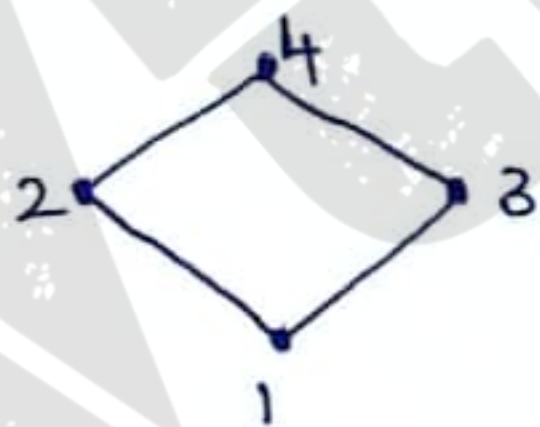
$\therefore R$  is antisymmetric.

iii) if  $(a,b) \in R$  and  $(b,c) \in R$  then we see that  $(a,c) \in R$ .

$\therefore R$  is transitive.

Thus  $R$  is a partial order on  $A$ .  $\therefore (A, R)$  is a Poset.

The Hasse Diagram for  $R$  is as shown below:



$\rightarrow$  NO  $\Delta^{\text{ie}}$  shud come in Hasse diagram

$\rightarrow$  2 no's which are related to each other shud not be in same line

2) Let  $R$  be a relation on the set  $A = \{1, 2, 3, 4\}$  defined by

Hw  $x R y$  iff  $x$  divides  $y$ . Prove that  $(A, R)$  is a Poset.

Draw its Hasse diagram.

Soln:-  $R = \{(x,y) \mid x, y \in A \text{ and } x \text{ divides } y\}$

$\therefore R = \{(1,1)(1,2)(1,3)(1,4)(2,2)(2,4)(3,3)(4,4)\}$ .

i) we see that  $(a,a) \in R \quad \forall a \in A$ .  $\Rightarrow R$  is reflexive on  $A$ .

ii) If  $(a,b) \in R$  and  $a \neq b$ , then we see that  $(b,a) \notin R$ .

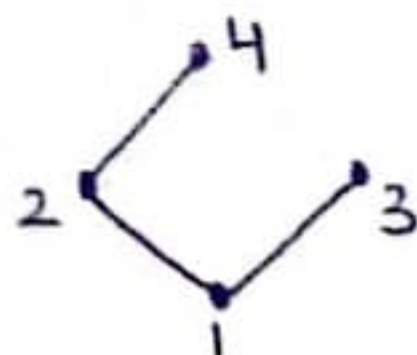
$\therefore R$  is antisymmetric on  $A$ .

iii) if  $(a, b) \in R$  and  $(b, c) \in R$ , then we see that  $(a, c) \in R$ .

$\therefore R$  is transitive.

Thus  $R$  is a partial order on  $A$ .  $\underline{(A, R)}$  is a Poset.

The Hasse diagram for  $R$  is as shown below:



3) Let  $A = \{1, 2, 3, 4, 6, 8, 12\}$ . on  $A$ , define the partial ordering relation  $R$  by  $xRy$  iff  $x|y$ . Prove that  $R$  is a partial order on  $A$ . Draw the Hasse diagram for  $R$ .

Sol<sup>n</sup>:  $R = \{(1, 1)(1, 2)(1, 3)(1, 4)(1, 6)(1, 8)(1, 12)(2, 2)(2, 4)(2, 6)(2, 8)(2, 12)(3, 3)(3, 6)(3, 12)(4, 4)(4, 8)(4, 12)(6, 6)(6, 12)(8, 8)(12, 12)\}$ .

i) we see that  $(a, a) \in R \forall a \in A \Rightarrow R$  is reflexive.

ii) if  $(a, b) \in R$  and  $a \neq b$ , then we see that  $(b, a) \notin R$ .

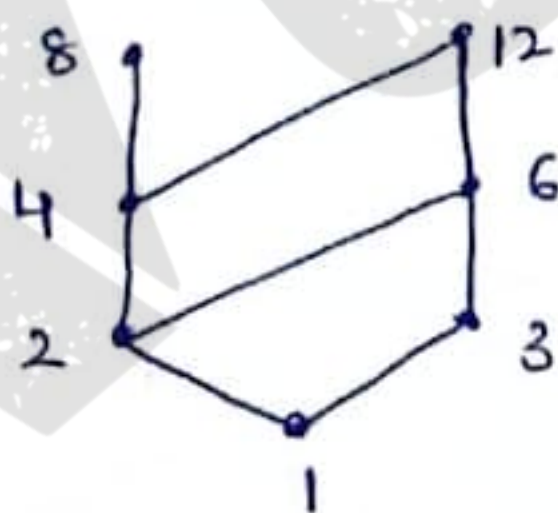
$\therefore R$  is antisymmetric.

iii) if  $(a, b) \in R$  and  $(b, c) \in R$ , then we see that  $(a, c) \in R$ .

$\therefore R$  is transitive.

Thus  $R$  is a partial order on  $A$ .  $\underline{(A, R)}$  is a Poset.

The Hasse diagram is as below:



4) Draw the Hasse diagram representing the positive divisors of 36.

Sol<sup>n</sup>: The set of all positive divisors of 36 are:

$$D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}.$$

The relation  $R$  of divisibility  $\underline{aRb}$  iff  $a$  divides  $b$ , is a partial order on this set.

we note that, under  $R$ ,

1 is related to all elements of  $D_{36}$ ,

2 is " — 2, 4, 6, 12, 18, 36

3 " — 3, 6, 9, 12, 18, 36

4 " — 4, 12, 36

6 " — 6, 12, 18, 36

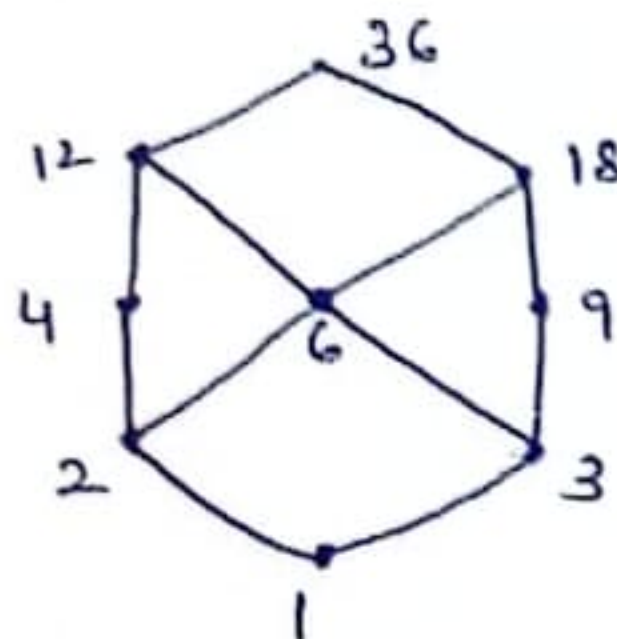
9 " — 9, 18, 36

12 " — " 12, 36

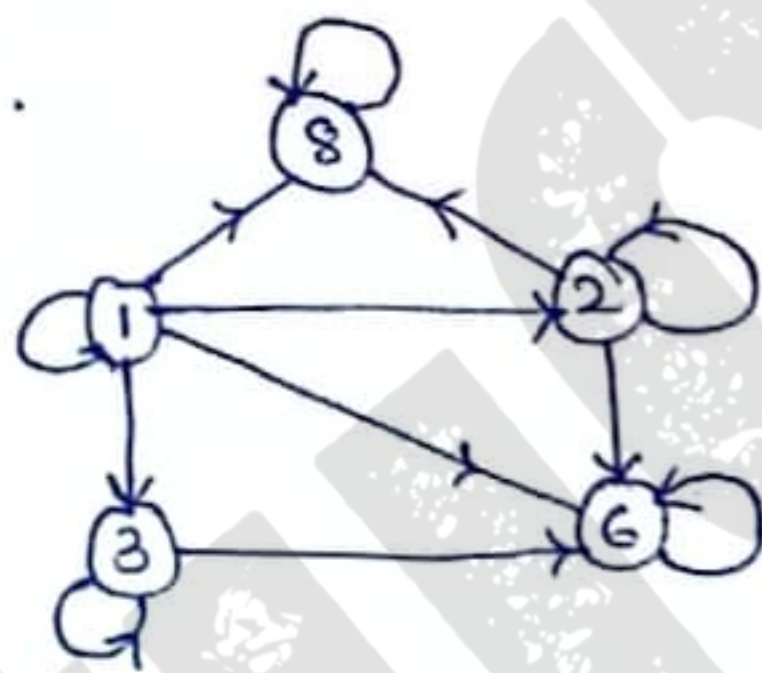
18 " — " 18, 36

36 " — 36.

Hasse diagram :-



5) The digraph for a relation on the set  $A = \{1, 2, 3, 6, 8\}$  is as shown below: Verify that  $(A, R)$  is a Poset and write down its Hasse diagram.



Soln:-  $R = \{(1,1)(1,2)(1,3)(1,6)(1,8)(2,2)(2,6)(2,8)(3,3)(3,6)(6,6)(8,8)\}$ .

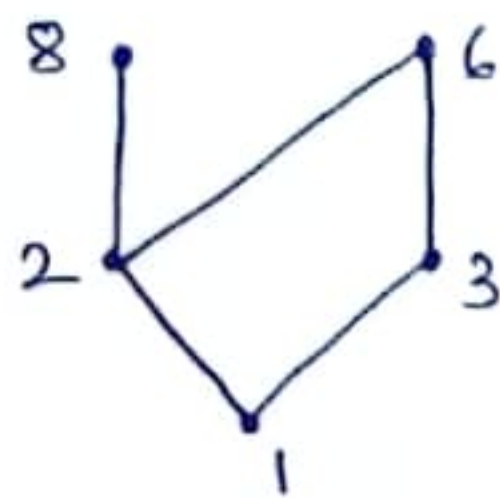
i)  $R$  is reflexive b'coz  $(a,a) \in R \forall a \in A$ .

ii)  $R$  is antisymmetric b'coz if  $(a,b) \in R$  and  $a \neq b$ , then we see that  $(b,a) \notin R$ .

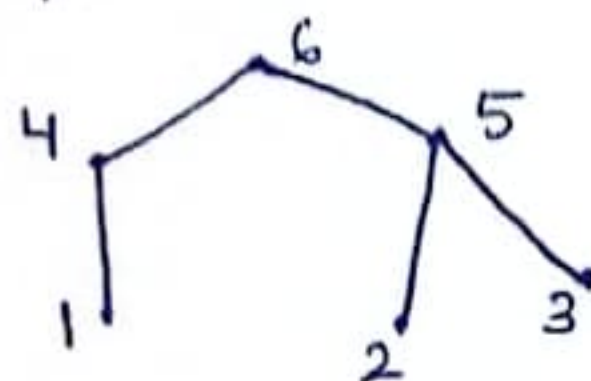
iii)  $R$  is transitive b'coz if  $(a,b) \in R$  and  $(b,c) \in R$ , we see that  $(a,c) \in R$ .

Thus  $R$  is a partial order on  $A$ ,  $\therefore (A, R)$  is a Poset.

Hasse diagram :



6) The Hasse diagram of a partial order  $R$  on the set  $A = \{1, 2, 3, 4, 5, 6\}$  is as given below. Write down  $R$  as a subset of  $A \times A$ , construct its digraph. Also determine the matrix of the Partial order.



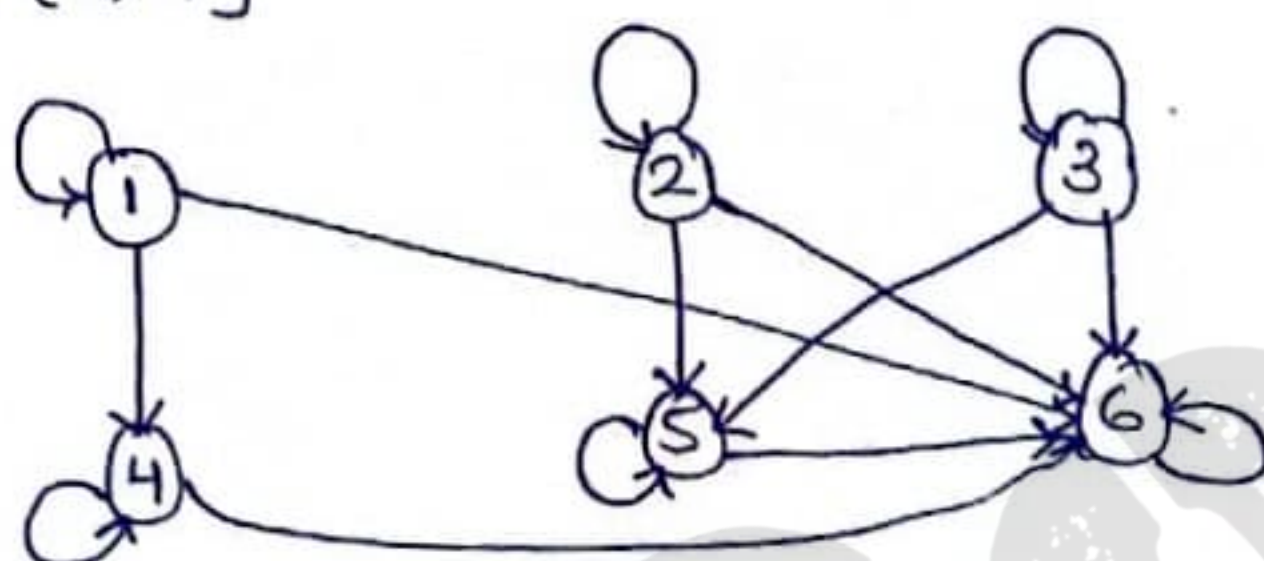
Soln:- From the diagram,

$1R1, 1R4, 1R6, 2R2, 2R5, 2R6,$

$3R3, 3R5, 3R6, 4R4, 4R6, 5R5, 5R6, 6R6.$

$\therefore R = \{(1,1) (1,4) (1,6) (2,2) (2,5) (2,6) (3,3) (3,5) (3,6) (4,4) (4,6) (5,5) (5,6) (6,6)\}$

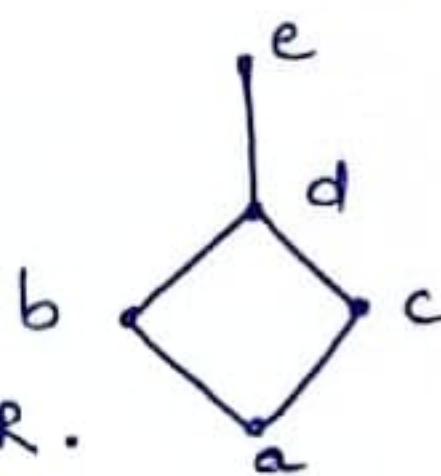
Digraph :-



$M(R) = M_R =$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 3 & 0 & 0 & 1 & 0 & 1 & 1 \\ 4 & 0 & 0 & 0 & 1 & 0 & 1 \\ 5 & 0 & 0 & 0 & 0 & 1 & 1 \\ 6 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

7) For  $A = \{a, b, c, d, e\}$ , the Hasse diagram for the Poset  $(A, R)$  is as shown below:



i) Determine the relation matrix for  $R$ .

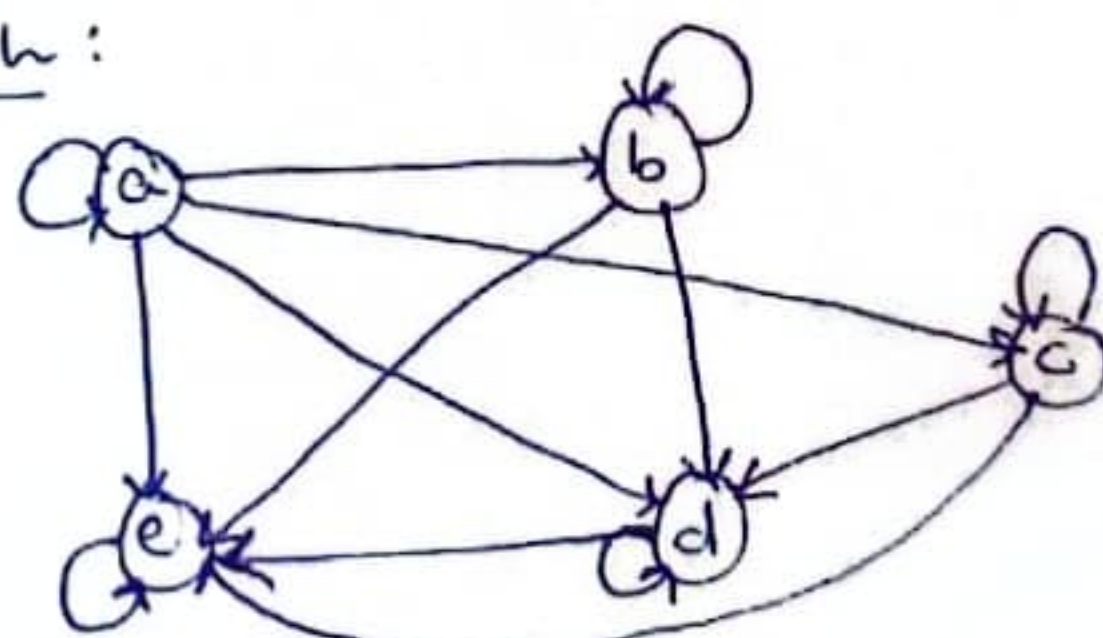
ii) Construct the digraph for  $R$ .

Soln:-  $R = \{(a,a) (a,b) (a,c) (a,d) (a,e) (b,b) (b,d) (b,e) (c,c) (c,d) (c,e) (d,d) (d,e) (e,e)\}$

i)  $M(R) =$

$$\begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

ii) Digraph :

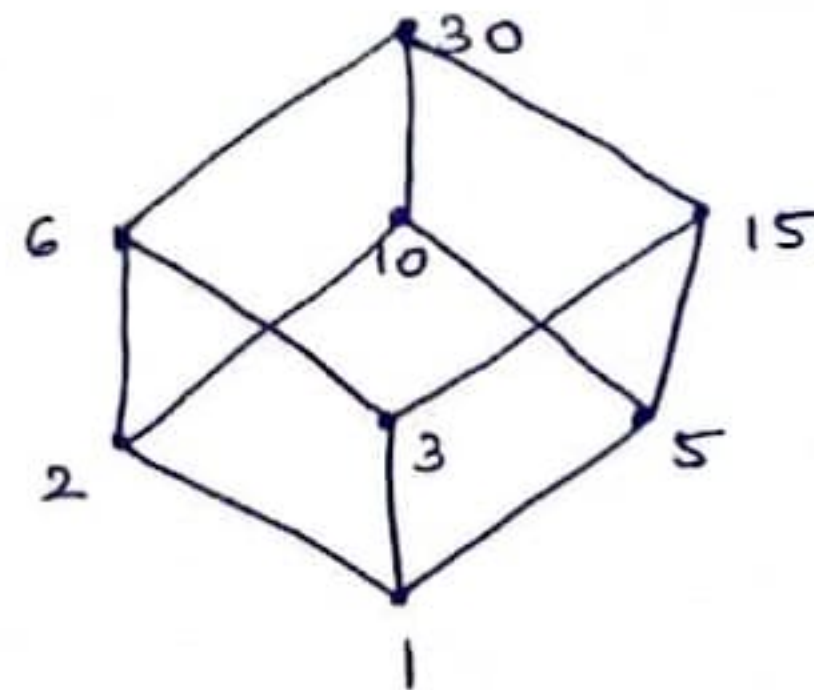


8) In the following cases, consider the partial order of divisibility on the set  $A$ . Draw the Hasse diagram for the Poset and determine whether the Poset is totally ordered or not.

(i)  $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$

(ii)  $A = \{2, 4, 8, 16, 32\}$ .

Soln:- i)  $R = \{(1,1) (1,2) (1,3) (1,5) (1,6) (1,10) (1,15) (1,30) (2,2) (2,6) (2,10) (2,30) (3,3) (3,6) (3,15) (3,30) (5,5) (5,10) (5,15) (5,30) (6,6) (6,30) (10,10) (10,30) (15,15) (15,30) (30,30)\}$



$(A, R)$  is not totally ordered set because if we consider 2, 3 in  $A$  neither 2 divides 3 nor 3 divides 2.

ii)  $R = \{(2,2) (2,4) (2,8) (2,16) (2,32) (4,4) (4,8) (4,16) (4,32) (8,8) (8,16) (8,32) (16,16) (16,32) (32,32)\}$ .



$(A, R)$  is totally ordered set.

Extremal elements in Poset : Consider a Poset  $(A, R)$ , we define some special elements called extremal elements that may exist in  $A$ .

- 1) An element  $a \in A$  is called a maximal element if there exists no  $x$  in  $A$  other than ' $a$ ' such that  $aRx$  i.e. ' $a$ ' is maximal element if and only if in <sup>bottom</sup> Hasse diagram of  $R$ , no edge starts at ' $a$ '.
- 2) An element  $a \in A$  is called a minimal element if there exists no  $x$  in  $A$  other than ' $a$ ' such that  $xRa$  i.e. ' $a$ ' is minimal element if and only if in Hasse diagram of  $R$ , no edge terminates at ' $a$ '.
- 3) An element  $a \in A$  is called a greatest element of  $A$  if  $xRa \quad \forall x \in A$ . i.e. all elements of  $A$  should be related to ' $a$ '.
- 4) An element  $a \in A$  is called a least element of  $A$  if  $aRx \quad \forall x \in A$  i.e. ' $a$ ' should be related to all elements of  $A$ .
- 5) Let  $B \subseteq A$ . Then an element  $a \in A$  is called an upper bound of  $B$  if  $xRa, \forall x \in B$ . i.e. all elements of subset  $B$  should be related to ' $a$ '.
- 6) Let  $B \subseteq A$ . Then an element  $a \in A$  is called a lower bound of  $B$  if  $aRx, \forall x \in B$ . i.e. ' $a$ ' should be related to all elements of subset  $B$ .
- 7) Let  $B \subseteq A$ . Then an element  $a \in A$  is called the Least Upper bound (LUB) of  $B$  if
  - i)  $a$  is an upper bound of  $B$  and
  - ii) If  $a'$  is <sup>also</sup> an upper bound of  $B$ , then  $aRa'$
- 8) Let  $B \subseteq A$ . Then an element  $a \in A$  is called the greatest lower bound (GLB) of  $B$  if
  - i)  $a$  is a lower bound of  $B$  and
  - ii) If  $a'$  is <sup>also</sup> a lower bound of  $B$ , then  $a'Ra$ .

Note:- LUB is also called Supremum and  
GLB is " ————— Infimum.

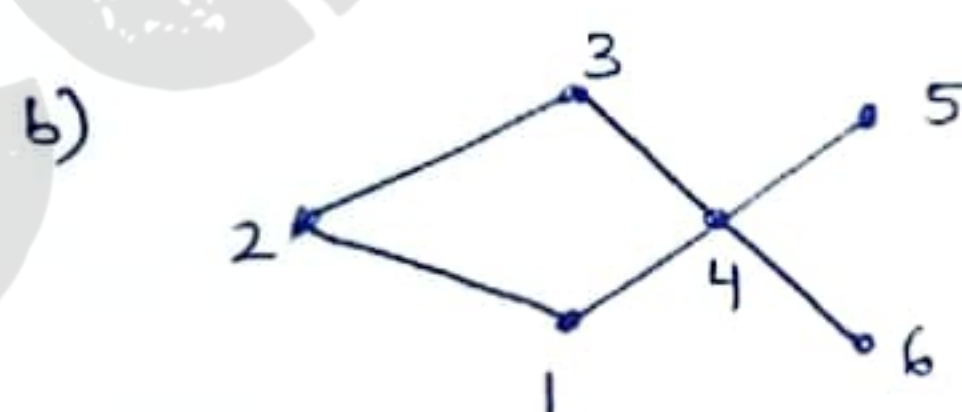
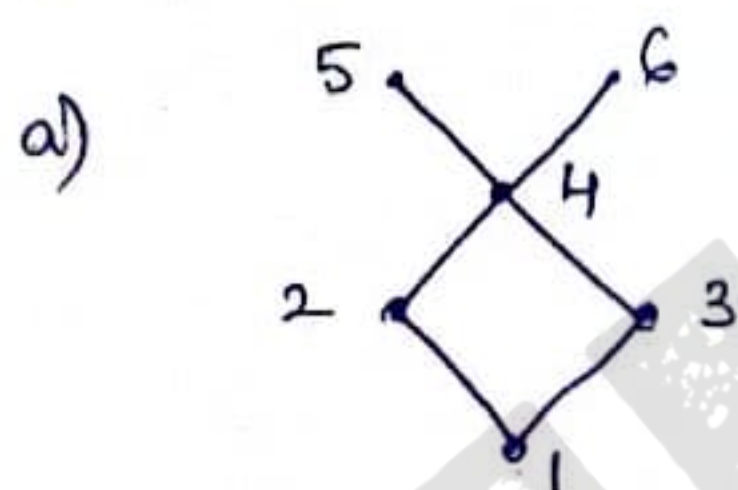
Lattices :- Let  $(A, R)$  be a Poset. This is called a Lattice if  $\forall x, y \in A$ , the elements  $LUB\{x, y\}$  and  $GLB\{x, y\}$  exist in  $A$ .

Ex :-  $(N, \leq)$  is a lattice. for all  $x, y$  in  $N$ ,  
 $GLB\{x, y\} = \min\{x, y\}$  and  $LUB\{x, y\} = \max\{x, y\}$ .  
 Both of these belong to  $N$ .

2)  $(Z^+, |)$  is a lattice where  $|$  is the divisibility relation.  
 for all  $x, y$  in  $Z^+$ ,  $GLB\{x, y\} = \gcd(x, y)$  and  
 $LUB\{x, y\} = \text{lcm}(x, y)$ . Both of these belong to  $Z^+$ .

### Problems :-

1) For the Poset  $(A, R)$  represented by the Hasse diagram, find  
 i) maximal ii) minimal iii) greatest and iv) least element(s)

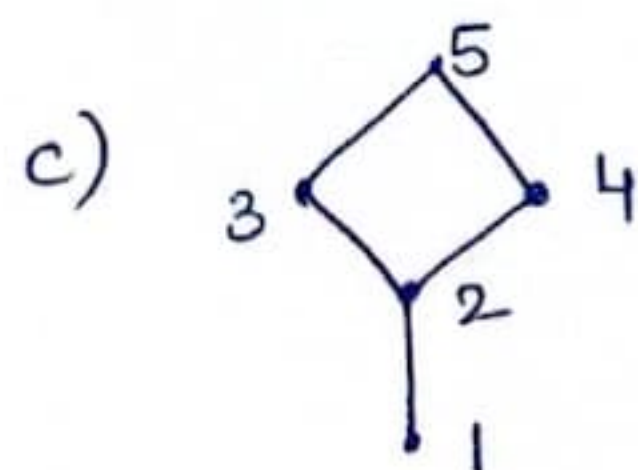


Soln :- a) 5 and 6 are maximal elements  
 1 is a minimal element.

1 is the least element ( $\because$  1 is related to all elements of  $A$ )  
 No greatest element ( $\because$  not  $5 \leq 6$  and not  $6 \leq 5$ )

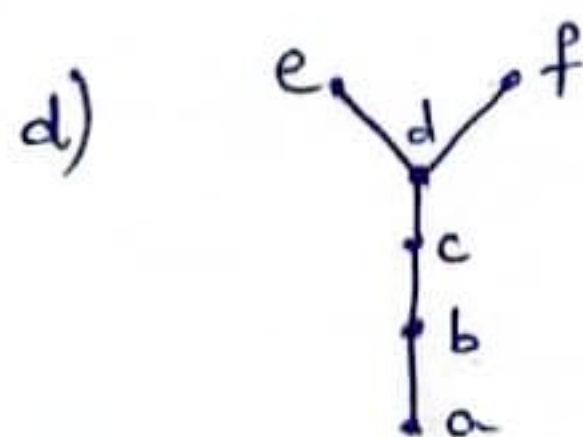
b) Maximal : 3, 5  
 Minimal : 1, 6

Greatest : No greatest ( $\because 5 R 3$  and  $3 R 5$ )  
 Least : No least ( $\because 1 R 6$  &  $6 R 1$ )



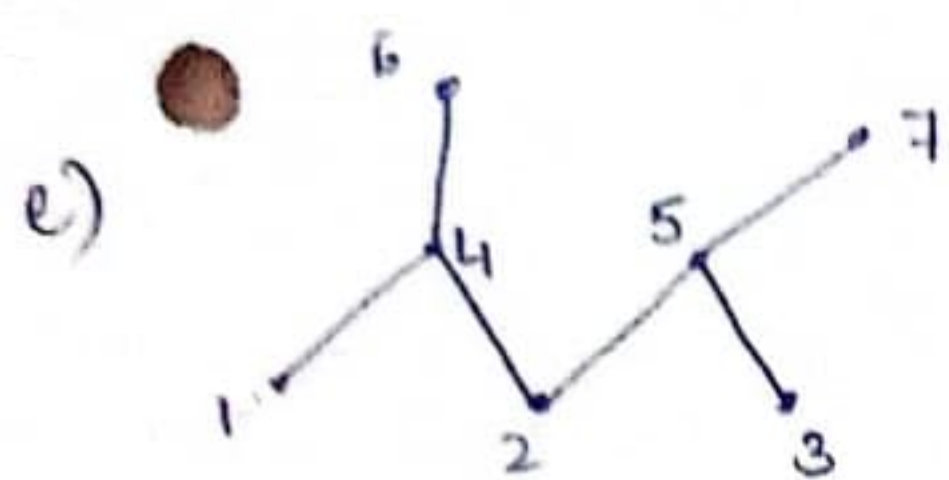
Maximal : 5  
 Minimal : 1

Greatest : 5 ( $\because 1 R 5, 2 R 5, 3 R 5, 4 R 5, 5 R 5$ )  
 Least : 1 ( $\because 1 R 1, 1 R 2, 1 R 3, 1 R 4, 1 R 5$ )



Maximal : e, f  
 Minimal : a

greatest : No  
 least : a

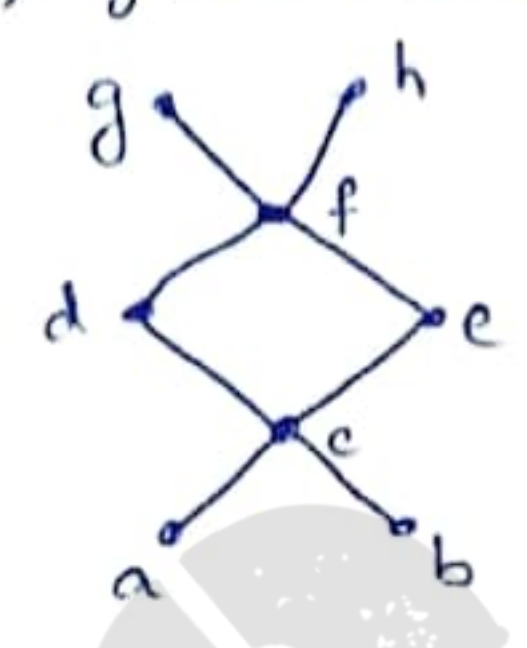


Maximal : 6, 7  
Minimal : 1, 2, 3

greatest : No  
least : No.

2) Consider the Hasse diagram of a Poset  $(A, R)$  given below.

If  $B = \{c, d, e\}$ , find (if they exist)



- i) all upper bounds of  $B$ .
- ii) all lower bounds of  $B$
- iii) the least upper bound of  $B$
- iv) the greatest lower bound of  $B$ .

Soln :- i) All of  $c, d, e$  which are in  $B$  are related to  $f, g, h$ .  
 $\therefore f, g, h$  are upper bounds of  $B$ .

ii) The elements  $a, b, c$  are related to all of  $c, d, e$  which are in  $B$ .  $\therefore a, b, c$  are lower bounds of  $B$ .

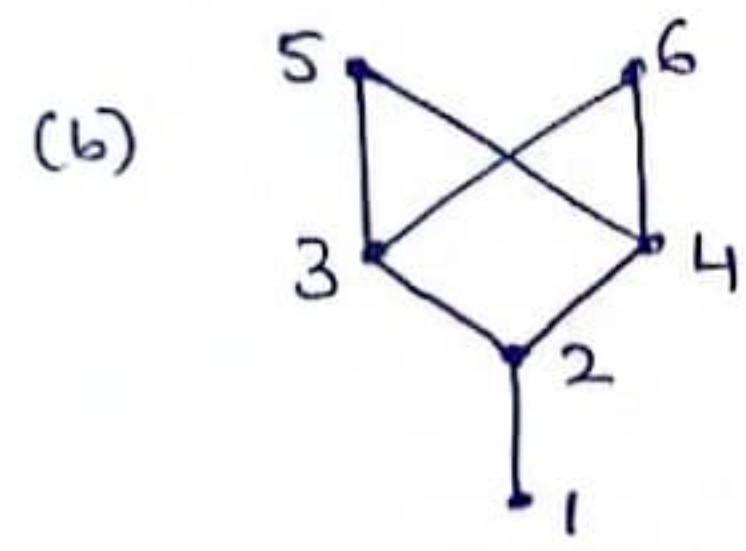
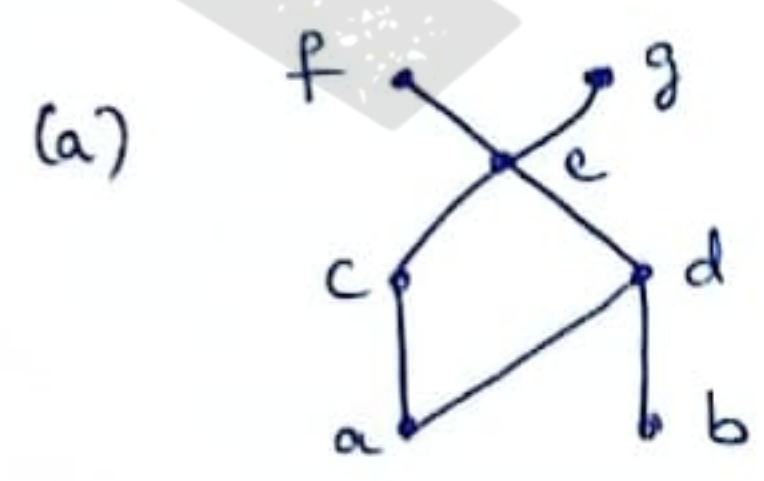
iii) The upper bound  $f$  of  $B$  is related to the other upper bounds  $g$  and  $h$  of  $B$ .  $\therefore f$  is the LUB of  $B$ .  $\underline{\underline{LUB(B) = f}}$

iv) The lower bounds  $a$  and  $b$  of  $B$  are related to the lower bound  $c$  of  $B$ .  $\therefore c$  is the GLB of  $B$ .  $\underline{\underline{GLB(B) = c}}$

3) For the Posets shown in the following Hasse diagrams, find

- i) all upper bounds
- ii) all lower bounds
- iii) LUB
- iv) GLB of the set  $B$ , where

$B = \{c, d, e\}$  in case (a) and  $B = \{3, 4, 5\}$  in case (b).



Soln :- (a) i) All of  $c, d, e$  which are in  $B$  are related to  $f, g$ .  
 $\therefore e, f, g$  are upper bounds of  $B$ .

ii) The element  $a$  is related to all of  $c, d, e$  which are in  $B$ .  $\therefore a$  is a lower bound of  $B$ .

ii) The upper bound  $e$  of  $B$  is related to the other upper bounds  $f$  and  $g$  of  $B$ .

∴  $e$  is the LUB of  $B$ .  $\therefore \text{LUB}(B) = e$ .

iv)  $\text{GLB}(B) = a$ .

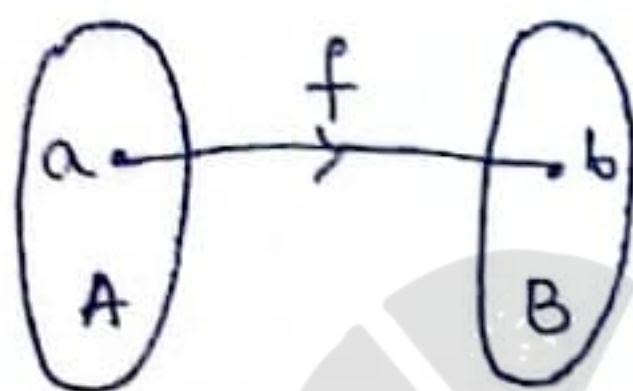
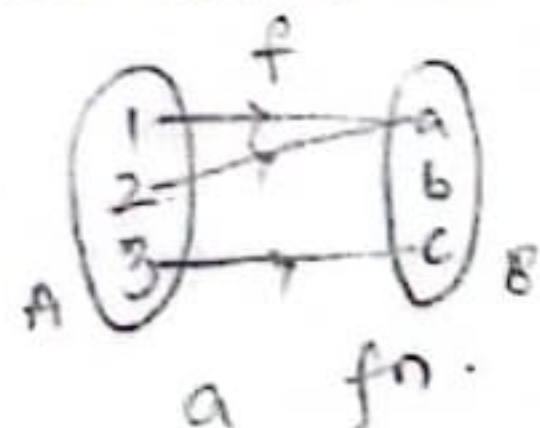
b) i)  $5$  is ~~an~~ the upper bound of  $B$ . (not  $6$  ∵  $5 \neq 6$ )

ii)  $1, 2$  are the lower bounds of  $B$ .

iii) There is no LUB of  $B$ .

iv)  $\text{GLB}(B) = 2$ .

Def<sup>n</sup>:- Let  $A$  and  $B$  be non empty sets. Then a function (or a mapping) from  $A$  to  $B$  is a relation from  $A$  to  $B$  such that for each  $a$  in  $A$ , there exists a unique  $b$  in  $B$  such that  $(a, b) \in f$ . Then we write  $b = f(a)$ . Here  $b$  is called the image of  $a$  and  $a$  is called the preimage of  $b$ , under  $f$ . The element  $a$  is <sup>also</sup> called an argument of the function  $f$  and  $b = f(a)$  is called the value of the function  $f$  for the argument  $a$ . A function  $f$  from  $A$  to  $B$  is denoted by  $f: A \rightarrow B$ . The Pictorial representation of  $f$  is as below:



"Every function is a relation but every relation is not a fn".

B'coz, if  $R$  is a relation from  $A$  to  $B$  then an element of  $A$  can be related to 2 elements of  $B$  under  $R$ .

But under a function, an element of  $A$  can be related to only one element of  $B$ .

For the function  $f: A \rightarrow B$ ,  $A$  is called the domain of  $f$  and  $B$  is called the co-domain of  $f$ . The subset of  $B$  consisting of the images of all elements of  $A$  under  $f$  is called the range of  $f$ , denoted by  $f(A)$ .

Note:-

- 1) Every  $a$  in  $A$  belong to some pair  $(a, b) \in f$  and if  $(a, b_1) \in f$  and  $(a, b_2) \in f$  then  $b_1 = b_2$ .
- 2) An element  $b \in B$  need not have a preimage in  $A$ , under  $f$ .
- 3) Two different elements of  $A$  can have the same image in  $B$ , under  $f$ .

4) The statements  $(a, b) \in f$ ,  $a f b$  and  $b = f(a)$  are equivalent.

5) If  $g$  is a function from  $A$  to  $B$ , then  $f = g$  iff  $f(a) = g(a)$ ,  $\forall a \in A$ .

6) The range of  $f: A \rightarrow B$  is given by  $f(A) = \{f(x) / x \in A\}$  and  $f(A)$  is a subset of  $B$ .

7) For  $f: A \rightarrow B$ , if  $A_1 \subseteq A$  and  $f(A_1)$  is defined by  $f(A_1) = \{f(x) / x \in A_1\}$ , then  $f(A_1) \subseteq f(A)$ . Here  $f(A_1)$  is called the image of  $A_1$ , under  $f$ .

8) For  $f: A \rightarrow B$ , if  $b \in B$  and  $f^{-1}(b)$  is defined by  $f^{-1}(b) = \{x \in A / f(x) = b\}$ , then  $f^{-1}(b) \subseteq A$ . Here  $f^{-1}(b)$  is called the preimage of  $b$ , under  $f$ .

9) For  $f: A \rightarrow B$ , if  $B_1 \subseteq B$  and  $f^{-1}(B_1)$  is defined by  $f^{-1}(B_1) = \{x \in A / f(x) \in B_1\}$ , then  $f^{-1}(B_1) \subseteq A$ . Here  $f^{-1}(B_1)$  is called the preimage of  $B_1$ , under  $f$ .

### Types of functions :-

1) Identity function :- A  $f^n$   $f$  on a set  $A$  is an identity function if the image of every element of  $A$  (under  $f$ ) is

itself and is denoted by  $I_A$ .

i.e.  $f: A \rightarrow A$  <sup>only, not  $A \rightarrow B$</sup>  is such that  $f(a) = a \quad \forall a \in A$ .

In case of identity function,  $f(A) = A$ .

2) Constant function :- A  $f^n$   $f: A \rightarrow B$  is called a constant  $f^n$  if  $f(a) = c \quad \forall a \in A$ . i.e.  $f$  is a constant  $f^n$  if image of every element of  $A$  is same in  $B$ , and in this case  $f(A) = \{c\}$ .

3) onto function :- <sup>(surjective fn)</sup>  $f: A \rightarrow B$  is said to be an onto fn from A to B if for every element b of B, there exists an element a of A such that  $f(a) = b$ .  
 i.e., f is an onto fn from A to B if every element b of B has a preimage in A. i.e. range of f = B.  
 No element in B should be free.

4) one-to-one function :- <sup>(Injective fn)</sup>  $f: A \rightarrow B$  is said to be an one-to-one fn from A to B if different elements of A have different images in B under f.  
 i.e. if  $f(a_1) = f(a_2)$  then  $a_1 = a_2$  where  $a_1, a_2 \in A$ .  
 (or) if  $a_1 \neq a_2$  then  $f(a_1) \neq f(a_2)$  (by contrapositive).

5) one-to-one correspondence <sup>(Bijective fn or Bijection)</sup> :-  
 $f: A \rightarrow B$  is said to be an one-to-one correspondence if f is both one-to-one and onto. If  $f: A \rightarrow B$  is a bijective fn, then every element of A has a unique image in B and every element of B has a unique preimage in A.

Properties of functions :-

1) Let  $f: X \rightarrow Y$  be a fn and A and B be arbitrary non-empty subsets of X, then

i) If  $A \subseteq B$ , then  $f(A) \subseteq f(B)$

ii)  $f(A \cup B) = f(A) \cup f(B)$ .

iii)  $f(A \cap B) \subseteq f(A) \cap f(B)$  and the equality holds if f is 1-1.

2) Let A and B be finite sets and f be a fn from A to B, then the following are true:

i) If f is one-to-one then  $|A| \leq |B|$ .

ii) If f is onto then  $|B| \leq |A|$ .

iii) If f is bijective then  $|A| = |B|$ .

3) Suppose  $|A| = |B|$  and  $f: A \rightarrow B$ , then f is one-to-one iff f is onto.

4) If  $f: A \rightarrow B$  and  $|A| = |B|$ , then  $f$  is bijective iff  $f$  is one-to-one or onto.

5) If  $f: A \rightarrow B$ ,  $|A| = m$  and  $|B| = n$  then there are  $n^m$  functions from  $A$  to  $B$  and if  $m \leq n$  then there are  $\frac{n!}{(n-m)!}$  one-to-one functions from  $A$  to  $B$ .

Stirling number of second kind :-

Let  $A$  and  $B$  be finite sets with  $|A| = m$  and  $|B| = n$ , where  $m \geq n$ . Then the number of onto functions from  $A$  to  $B$  is given by the formula :

$$p(m, n) = \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m$$

(If  $m < n$ , then there are no onto fn from  $A$  to  $B$ )  
with  $p(m, n)$  given by the above formula, the no.  $\left[ \frac{p(m, n)}{n!} \right]$  is called the Stirling number of the second kind and is denoted by  $S(m, n)$ .

i) The no. of possible ways to assign ' $m$ ' distinct objects to ' $n$ ' identical places (containers) with no place (container) left empty is given by

$$S(m, n) = \frac{p(m, n)}{n!} = \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m, \text{ for } m \geq 1$$

ii) The no. of possible ways to assign ' $m$ ' distinct objects to ' $n$ ' identical places with empty places allowed is given by

$$p(m) = \sum_{i=1}^n S(m, i), \text{ for } m \geq n.$$

Problems :

1) Let  $A = \{1, 2, 3, 4\}$ . Determine whether or not the following relations on  $A$  are fns :

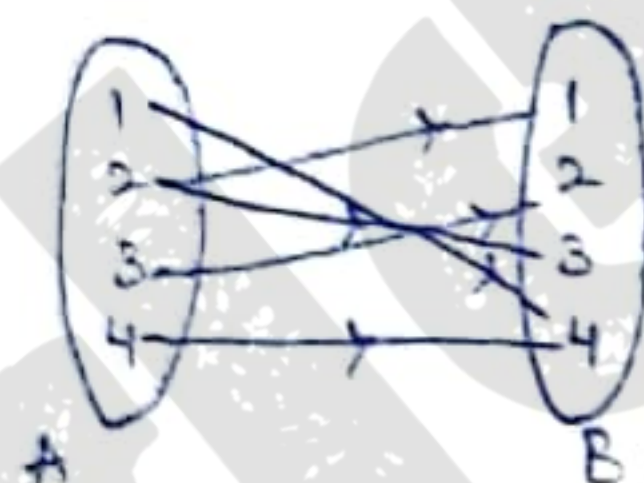
i)  $f = \{(2, 3) (1, 4) (2, 1) (3, 2) (4, 4)\}$

ii)  $g = \{(2, 1) (3, 4) (1, 4) (2, 1) (4, 4)\}$

Soln :-

i) we see that  $(2, 3) \in f$  and  $(2, 1) \in f$  i.e. the element 2 is related to two different elements 3 and 1, under  $f$ .

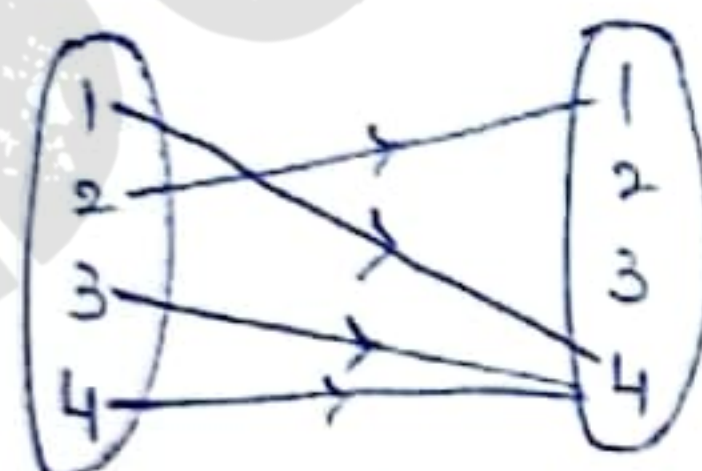
$\therefore f$  is not a fn.



ii) we see that, under  $g$ , every element of  $A$  is related to a unique element of  $A$ .

$\therefore g$  is a fn from  $A$  to  $A$ .

$$\text{Range of } g = g(A) = \{1, 4\}$$



(2, 1) appears twice. This has no special significance)

2) Let  $A = \{0, \pm 1, \pm 2, \pm 3\}$ . Consider  $f: A \rightarrow \mathbb{R}$  (where  $\mathbb{R}$  is the set of all real numbers) defined by  $f(x) = x^3 - 2x^2 + 3x + 1$  for  $x \in A$ . Find the range of  $f$ .

Soln :-  $f(0) = 1$ .

$$f(1) = 1^3 - 2(1)^2 + 3(1) + 1 = 3$$

$$f(2) = 7, \quad f(3) = 19, \quad f(-1) = -5, \quad f(-2) = -21, \quad f(-3) = -53.$$

$$\therefore \text{Range of } f = \{-53, -21, -5, 1, 3, 7, 19\}.$$

3) Let  $A = \{1, 2, 3, 4, 5, 6\}$  and  $B = \{6, 7, 8, 9, 10\}$ . If a fn  $f: A \rightarrow B$  is defined by  $f = \{(1, 7) (2, 7) (3, 8) (4, 6) (5, 9) (6, 9)\}$ .

Determine  $f^{-1}(6)$  and  $f^{-1}(9)$ . If  $B_1 = \{7, 8\}$  and  $B_2 = \{8, 9, 10\}$

find  $f^{-1}(B_1)$  and  $f^{-1}(B_2)$ .

Soln:- By def<sup>n</sup>,

$$f^{-1}(b) = \{x \in A \mid f(x) = b\}$$

$$\therefore f^{-1}(6) = \{x \in A \mid f(x) = 6\} = \{4\}.$$

$$f^{-1}(9) = \{x \in A \mid f(x) = 9\} = \{5, 6\}.$$

For any  $B_1 \subseteq B$ , by def<sup>n</sup>

$$f^{-1}(B_1) = \{x \in A \mid f(x) \in B_1\}.$$

For  $B_1 = \{7, 8\}$ ,  $f(x) \in B_1$  when  $f(x) = 7$  and  $f(x) = 8$ .  
From the def<sup>n</sup> of  $f$ , we see that  $f(x) = 7$  when  $x = 1$  &  $x = 2$ ,  
and  $f(x) = 8$  when  $x = 3$ .

$$\therefore f^{-1}(B_1) = \{1, 2, 3\}.$$

$$\text{iii) } f^{-1}(B_2) = \{x \in A \mid f(x) \in B_2\} = \{3, 5, 6\}.$$

4) Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \begin{cases} 3x-5 & \text{for } x > 0 \\ -3x+1 & \text{for } x \leq 0 \end{cases}$ .

i) Determine  $f(0)$ ,  $f(-1)$ ,  $f(\frac{5}{3})$ ,  $f(-\frac{5}{3})$

ii) Find  $f^{-1}(0)$ ,  $f^{-1}(1)$ ,  $f^{-1}(-1)$ ,  $f^{-1}(3)$ ,  $f^{-1}(-3)$ ,  $f^{-1}(-6)$ .

iii) what are  $f^{-1}([-5, 5])$  and  $f^{-1}([-6, 5])$ ?

Soln:- i)  $f(0) = -3(0)+1 = 1$ .

$$f(-1) = -3(-1)+1 = 4.$$

$$f(\frac{5}{3}) = 3(\frac{5}{3})-5 = 0.$$

$$f(-\frac{5}{3}) = -3(-\frac{5}{3})+1 = 6.$$

ii) By def<sup>n</sup>  $f^{-1}(b) = \{x \in \mathbb{R} \mid f(x) = b\}$ .

a)  $f^{-1}(0) = \{x \in \mathbb{R} \mid f(x) = 0\}.$

Consider  $f(x) = 0$ .

$$3x-5 = 0 \quad (x > 0)$$

$$3x = 5$$

$$x = \frac{5}{3} > 0$$

possible.

$$\text{and } -3x+1 = 0 \quad (x \leq 0)$$

$$3x = 1$$

$$x = \frac{1}{3} > 0 \nmid \text{ not possible.}$$

Thus  $f^{-1}(0) = \{5/3\}$ .

b)  $f^{-1}(1) = \{x \in \mathbb{R} \mid f(x) = 1\}$

$$f(x) = 1 \Rightarrow 3x - 5 = 1 \quad (x > 0) \quad \text{and} \quad -3x + 1 = 1 \quad (x \leq 0)$$

$$\Rightarrow 3x = 6 \quad \text{and} \quad 3x = 0$$

$$x = 2 > 0 \quad \checkmark \quad \quad \quad x = 0 \quad \checkmark$$

$\therefore f^{-1}(1) = \{0, 2\}$ .

c)  $f^{-1}(-1) = \{x \in \mathbb{R} \mid f(x) = -1\}$

$$f(x) = -1 \Rightarrow 3x - 5 = -1 \quad (x > 0) \quad \text{and} \quad -3x + 1 = -1 \quad (x \leq 0)$$

$$\Rightarrow 3x = 4 \quad \text{and} \quad -3x = -2$$

$$x = 4/3 \quad \checkmark \quad \quad \quad x = 2/3 \quad \times$$

$\therefore f^{-1}(-1) = \{4/3\}$ .

d)  $f^{-1}(3) = \{x \in \mathbb{R} \mid f(x) = 3\}$

$$f(x) = 3 \Rightarrow 3x - 5 = 3 \quad (x > 0) \quad \text{and} \quad -3x + 1 = 3 \quad (x \leq 0)$$

$$3x = 8 \quad \text{and} \quad -3x = 2$$

$$x = 8/3 \quad \checkmark \quad \quad \quad x = -2/3 \quad \checkmark$$

$\therefore f^{-1}(3) = \{-2/3, 8/3\}$ .

e)  $f^{-1}(-3) = \{x \in \mathbb{R} \mid f(x) = -3\}$

$$f(x) = -3 \Rightarrow 3x - 5 = -3 \quad (x > 0) \quad \text{and} \quad -3x + 1 = -3 \quad (x \leq 0)$$

$$\Rightarrow 3x = 2 \quad \text{and} \quad -3x = -4$$

$$x = 2/3 \quad \checkmark \quad \quad \quad x = 4/3 \quad \times$$

$\therefore f^{-1}(-3) = \{2/3\}$

f)  $f^{-1}(-6) = \{x \in \mathbb{R} \mid f(x) = -6\}$

$$f(x) = -6 \Rightarrow 3x - 5 = -6 \quad (x > 0) \quad \text{and} \quad -3x + 1 = -6 \quad (x \leq 0)$$

$$\Rightarrow 3x = -1 \quad \text{and} \quad -3x = -7$$

$$x = -1/3 \quad \times \quad \quad \quad x = 7/3 \quad \times$$

$\therefore f^{-1}(-6) = \{\} = \phi$ .

$$\text{iii) a) } f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid f(x) \in [-5, 5]\} \\ = \{x \in \mathbb{R} \mid -5 \leq f(x) \leq 5\}.$$

$$\text{for } f(x) = 3x - 5 \quad (x > 0)$$

$$-5 \leq 3x - 5 \leq 5.$$

add 5 throughout

$$0 \leq 3x \leq 10.$$

$\div$  by 3.

$$0 \leq x \leq \frac{10}{3}$$

$$\text{for } f(x) = -3x + 1 \quad (x \leq 0)$$

$$-5 \leq -3x + 1 \leq 5.$$

add -1 throughout

$$-6 \leq -3x \leq 4.$$

$\div$  by 3.

$$-2 \leq -x \leq \frac{4}{3}.$$

$\times$ ply by -1

$$2 \geq x \geq -\frac{4}{3}$$

$$\underline{\underline{\text{i.e.}}} \quad -\frac{4}{3} \leq x \leq 2$$

$$\text{Combining, } f^{-1}([-5, 5]) = \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq 2 \text{ or } 0 \leq x \leq \frac{10}{3}\}$$

$$= \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq \frac{10}{3}\}$$

$$= \left[-\frac{4}{3}, \frac{10}{3}\right] \quad \hookrightarrow x \in \left[-\frac{4}{3}, \frac{10}{3}\right].$$

$$\text{b) } f^{-1}([-6, 5]) = \{x \in \mathbb{R} \mid f(x) \in [-6, 5]\}$$

$$= \{x \in \mathbb{R} \mid -6 \leq f(x) \leq 5\}.$$

$$\text{for } f(x) = 3x - 5 \quad (x > 0)$$

$$-6 \leq 3x - 5 \leq 5$$

add 5

$$-1 \leq 3x \leq 10.$$

$\div$  3

$$-\frac{1}{3} \leq x \leq \frac{10}{3}$$

$$\text{for } f(x) = -3x + 1 \quad (x \leq 0)$$

$$-6 \leq -3x + 1 \leq 5$$

add -1

$$-7 \leq -3x \leq 4$$

$\div$  3

$$-\frac{7}{3} \leq -x \leq \frac{4}{3}$$

$\times$ ply by -1.

$$\frac{7}{3} \geq x \geq -\frac{4}{3}$$

$$\underline{\underline{\text{i.e.}}} \quad -\frac{4}{3} \leq x \leq \frac{7}{3}.$$

$$\text{combining, } f^{-1}([-6, 5]) = \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq \frac{7}{3} \text{ or } -\frac{1}{3} \leq x \leq \frac{10}{3}\}$$

$$= \{x \in \mathbb{R} \mid -\frac{4}{3} \leq x \leq \frac{10}{3}\}$$

$$= \left[-\frac{4}{3}, \frac{10}{3}\right].$$

5) a) Let  $A$  and  $B$  be finite sets with  $|A| = m$  and  $|B| = n$ .

Find how many functions are possible from  $A$  to  $B$ ?

b) If there are 2187 functions from  $A$  to  $B$  and  $|B| = 3$ ,

what is  $|A|$ ?

Soln:- a) Let  $A = \{a_1, a_2, \dots, a_m\}$  and  $B = \{b_1, b_2, \dots, b_n\}$ . then

a fn  $f: A \rightarrow B$  is of the form

$$f = \{(a_1, x), (a_2, x), \dots, (a_m, x)\}, \text{ where } x \text{ stands for}$$

$b_j$  for some  $j$ .

Since there are ' $n$ ' no. of  $b_j$ 's, there are ' $n$ ' choices for  $x$  in each of the ' $m$ ' ordered pairs belonging to  $f$ .

$\therefore$  Total no. of choices for  $x$  is

$$n \times n \times \dots \times n \text{ (m factors)} = n^m.$$

Thus there are  $n^m = |B|^{|A|}$  possible fns from  $A$  to  $B$ .

b) Given  $|B| = n = 3$ , and  $n^m = 2187$ .

$$\therefore 3^m = 2187$$

$$\Rightarrow m \log_e 3 = \log_e 2187$$

$$\Rightarrow m = |A| = 7$$

6) Let  $f: A \rightarrow B$  be a function. Let  $C$  and  $D$  be arbitrary nonempty subsets of  $B$ . Prove the following:

$$i) f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D) \quad ii) f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

$$iii) f^{-1}(\bar{C}) = \overline{f^{-1}(C)}.$$

Soln:- i) for any  $x \in A$ ,

$$x \in f^{-1}(C \cup D) \Leftrightarrow f(x) \in \{C \cup D\}$$

$$\Leftrightarrow f(x) \in C \text{ or } f(x) \in D.$$

$$\Leftrightarrow x \in f^{-1}(C) \text{ or } x \in f^{-1}(D)$$

$$\Leftrightarrow x \in \{f^{-1}(C) \cup f^{-1}(D)\}$$

$$\therefore f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D).$$

ii) for any  $x \in A$ ,

$$x \in f^{-1}(C \cap D) \Leftrightarrow f(x) \in \{C \cap D\}$$

$$\Leftrightarrow f(x) \in C \text{ and } f(x) \in D.$$

$$\Leftrightarrow x \in f^{-1}(C) \text{ and } x \in f^{-1}(D)$$

$$\Leftrightarrow x \in \{f^{-1}(C) \cap f^{-1}(D)\}$$

$$\therefore f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

iii) for any  $x \in A$ ,

$$x \in f^{-1}(\overline{C}) \Leftrightarrow f(x) \in \overline{C}$$

$$\Leftrightarrow f(x) \notin C$$

$$\Leftrightarrow x \notin f^{-1}(C).$$

$$\Leftrightarrow x \in \overline{f^{-1}(C)}.$$

$$\therefore f^{-1}(\overline{C}) = \overline{f^{-1}(C)}.$$

7) How many 1-1 functions are possible from  $A \rightarrow B$  where  $|A| = m$  &  $|B| = n$ , If there are 60 1-1 functions from  $A \rightarrow B$  and  $|A| = 3$ ,  $|B| = ?$ .

soln:- No. of 1-1 functions from  $A$  to  $B$  is  $\frac{n!}{(n-m)!}$

$$\text{Given } \frac{n!}{(n-m)!} = 60, \quad |A| = m = 3, \quad |B| = n = ?$$

$$\frac{n!}{(n-3)!} = 60.$$

$$\Rightarrow \frac{n \times (n-1) \times (n-2)!}{(n-3)!} = 60$$

$$\Rightarrow n(n-1)(n-2) = 60.$$

$$\Rightarrow 5 \times 4 \times 3 = 60$$

$$\Rightarrow \boxed{n=5}$$

$$\underline{\underline{\therefore |B| = n = 5.}}$$

8) Find the nature of the following fns defined on  $A = \{1, 2, 3\}$ .

i)  $f = \{(1, 1) (2, 2) (3, 3)\}$     ii)  $g = \{(1, 2) (2, 2) (3, 2)\}$ .

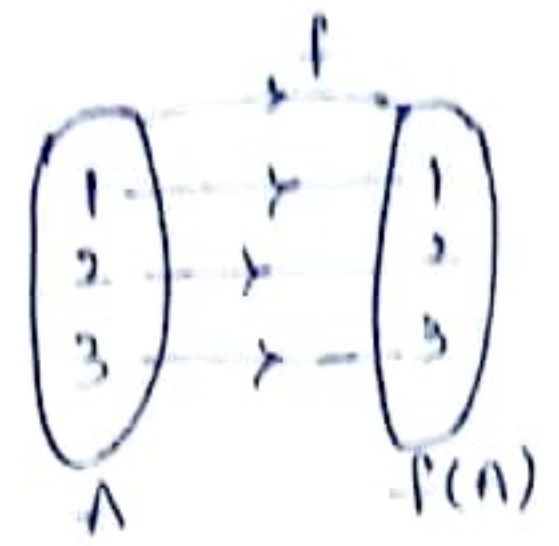
iii)  $h = \{(1, 2) (2, 3) (3, 1)\}$ .

Soln:- i) for every  $a \in A$ ,  $(a, a) \in f$  is a fa.

ie  $A = f(A)$

ie image of every element in  $A$  is itself.

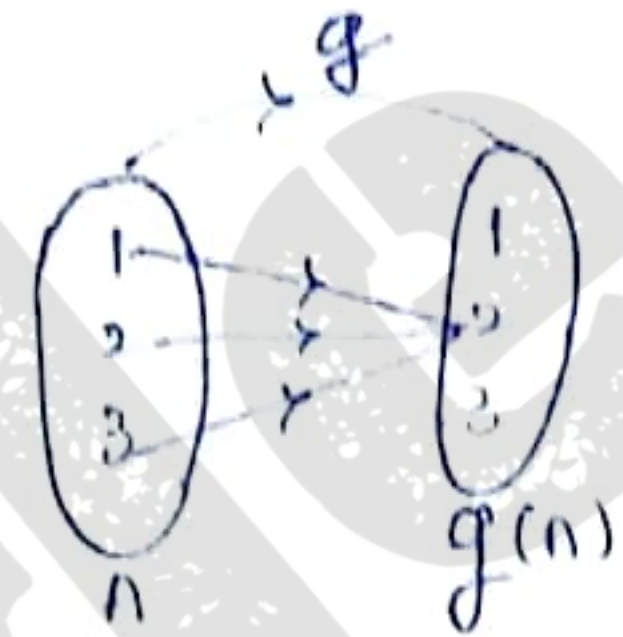
$\therefore f$  is the identity fn on  $A$ .



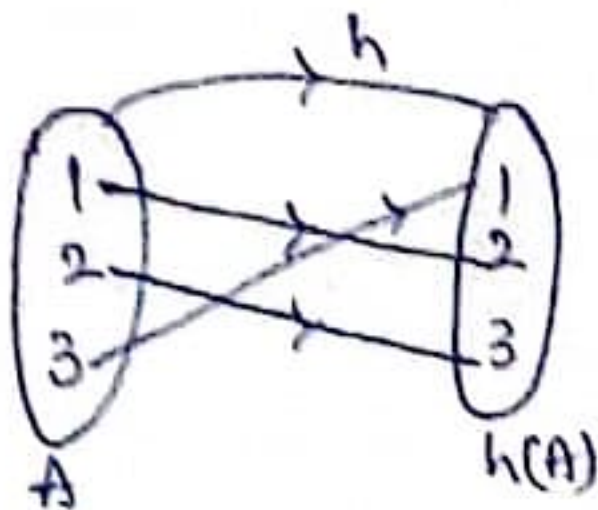
ii) we see that every  $a \in A$  has 2 as its image.

ie  $g(1) = 2, g(2) = 2, g(3) = 2$

$\therefore g$  is a constant fn on  $A$ .



iii)



we see that, every element of  $A$  has a unique image and every element of  $A$  has a unique preimage, under  $h$ .

$\therefore h$  is both one-to-one and onto.

$\therefore h$  is 1-1 correspondence.

9) Find whether the following fns from  $A$  to  $B$  are 1-1, onto:

i)  $f_1 = \{(1, 1) (2, 3) (3, 4)\}$

ii)  $f_2 = \{(1, 1) (2, 3) (3, 4) (4, 2)\}$

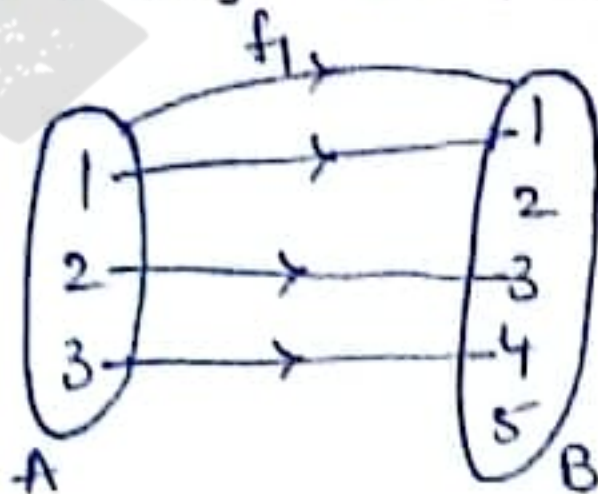
for  $A = \{1, 2, 3\}$ ,  $B = \{1, 2, 3, 4, 5\}$ .

for  $A = B = \{1, 2, 3, 4\}$

iii)  $A = \{a, b, c\}$   $B = \{1, 2, 3, 4\}$ ,  $f_3 = \{(a, 1) (b, 1) (c, 4)\}$ .

iv)  $A = \{1, 2, 3, 4\}$   $B = \{a, b, c, d\}$ ,  $f_4 = \{(1, a) (2, a) (3, d) (4, c)\}$ .

Soln:- i)

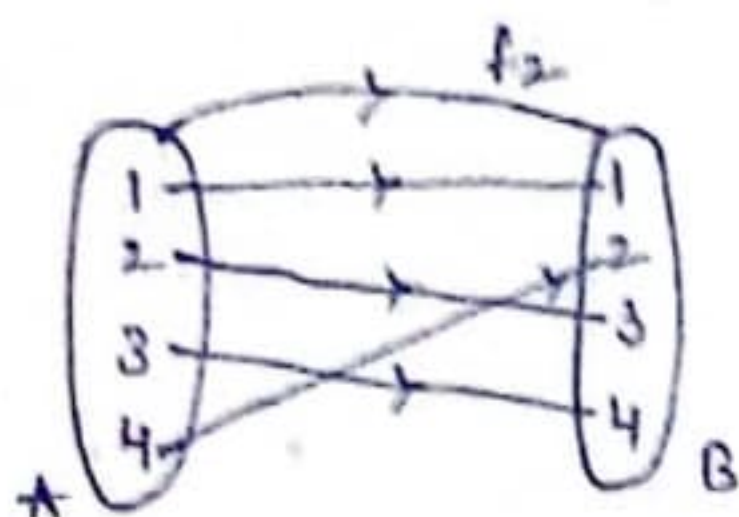


under  $f_1$ , every element of  $A$  has a unique image in  $B$  and no two elements of  $A$  have the same image in  $B$ .

$\therefore f$  is one-to-one.

under  $f_1$ , the elements 2 & 5 of  $B$  has no pre-image in  $A$ .  $\therefore f$  is not onto.

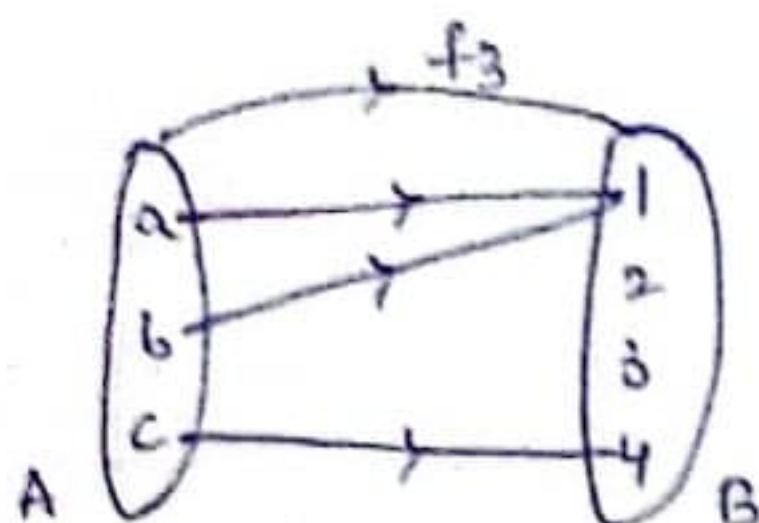
ii)



$f_2$  is both 1-1 & onto

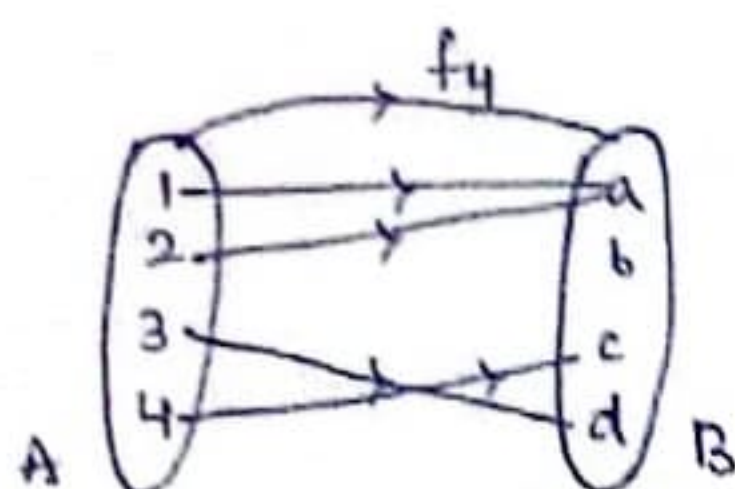
$\therefore f_2$  is bijective.

iii)



$f_3$  is neither 1-1 nor onto.

iv)



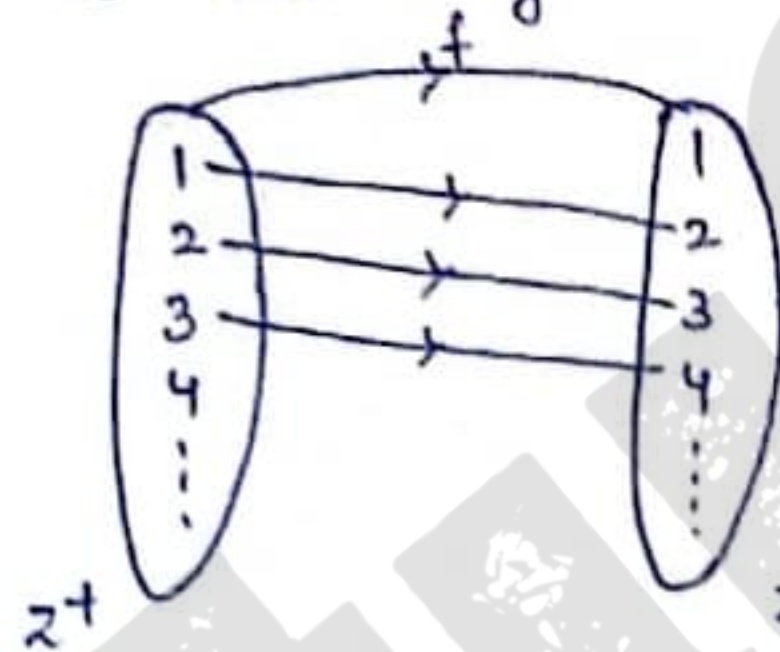
$f_4$  is neither 1-1 nor onto.

10) Let  $f, g: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$  where  $\forall x \in \mathbb{Z}^+, f(x) = x+1, g(x) = \max(1, x-1)$

i) what is the range of  $f$ ? ii) Is  $f$  1-1? iii) Is  $f$  onto?

iv) what is the range of  $g$ ? v) Is  $g$  1-1? vi) Is  $g$  onto?

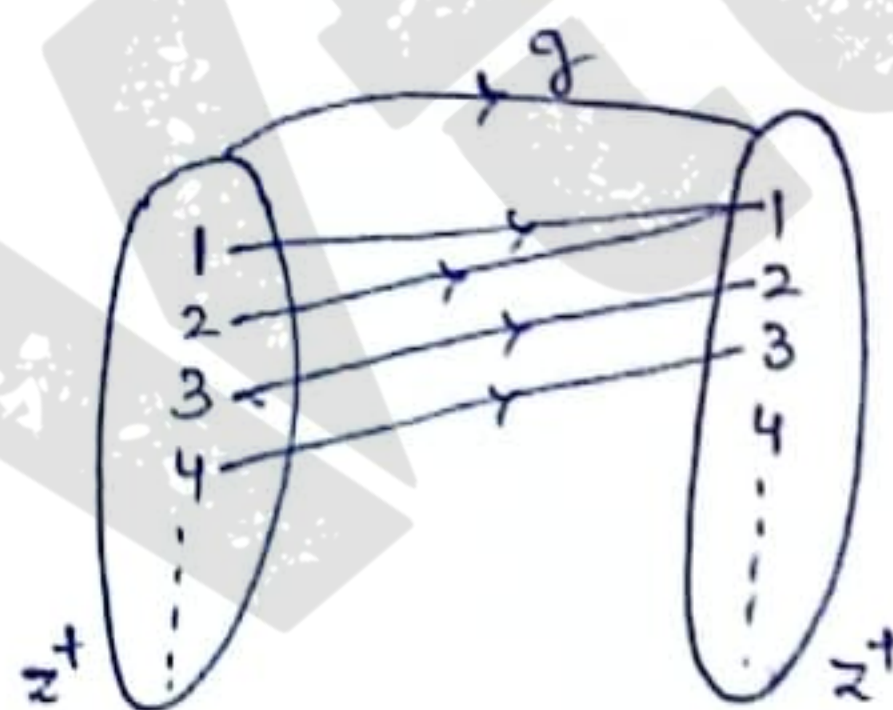
Soln:-



i) Range of  $f = \{2, 3, 4, \dots\}$   
 $= \mathbb{Z}^+ - \{1\}$

ii)  $f$  is 1-1

iii)  $f$  is not onto  $\because f^{-1}(1)$  does not exist.



$g(x) = \max(1, x-1)$

$g(1) = \max(1, 0) = 1$

$g(2) = \max(1, 1) = 1$

$g(3) = \max(1, 2) = 2$

$g(4) = \max(1, 3) = 3$  etc

iv) Range of  $g = \{1, 2, 3, \dots\} = \mathbb{Z}^+$

v)  $g$  is not 1-1

vi)  $g$  is onto.

Do 15th after this

P.T.O.

1) Let  $A = \{1, 2, 3, 4\}$  and  $B = \{1, 2, 3, 4, 5, 6\}$ .

i) Find how many fns are there from A to B. How many of these are 1-1? How many are onto?

ii) Find how many fns are there from B to A. How many of these are 1-1? How many are onto?

Soln :- Here  $|A| = m = 4$  and  $|B| = n = 6$ .

i) The no. of fns possible from A to B is  $n^m = 6^4 = 1296$ .

The no. of 1-1 fns possible from A to B is

$$\frac{n!}{(n-m)!} = \frac{6!}{2!} = 360.$$

There is no onto fn from A to B. ( $\because m < n$ )

ii) The no. of fns possible from B to A is  $m^n = 4^6 = 4096$ .

There is no 1-1 fn from B to A ( $\because m \neq n$ ).

The no. of onto fns from B to A is

$$p(n, m) = \sum_{k=0}^m (-1)^k \binom{m}{m-k} (m-k)^n$$

$$p(6, 4) = \sum_{k=0}^4 (-1)^k \binom{4}{4-k} (4-k)^6$$

$$= {}^4C_4 \times 4^6 - {}^4C_3 \times 3^6 + {}^4C_2 \times 2^6 - {}^4C_1 \times 1^6$$

$$= 1560.$$

Note :- 1) 1-1  $\rightarrow \frac{n!}{(n-m)!}$   
for  $m \leq n$

2) onto  $\rightarrow p(m, n)$  for  $m \geq n$ .

12) There are 6 Programmers who can assist eight executives.

In how many ways can the executives be assisted so that each programmer assists atleast one executive?

Soln :- all programmer shud have an executive.

use fn def<sup>n</sup> concept  
no 'a' in A shud be free

Let A denote the set of executives and B denote the set of programmers.  $\therefore |A| = m = 8$ ,  $|B| = n = 6$ .

P.T.O.

Thus Required no. = no. of onto fns from A to B.

$$\Rightarrow p(m, n) = p(8, 6) = (6!) \times S(8, 6).$$

$$\therefore S(8, 6) = \frac{1}{6!} \sum_{k=0}^6 (-1)^k \binom{6}{6-k} (6-k)^8$$

$$= \frac{1}{6!} \{ {}^6C_0 \times 6^8 - {}^6C_1 \times 5^8 + {}^6C_2 \times 4^8 - {}^6C_3 \times 3^8 + {}^6C_4 \times 2^8 - {}^6C_5 \times 1^8 \}$$

$$= 266.$$

$$\therefore p(8, 6) = 6! \times 266 = 191520.$$

13) Find the no. of ways of distributing 4 distinct objects among 3 identical containers, with some container(s) possible empty.

Soln:- Here, no. of objects is  $m=4$  and no. of containers is  $n=3$ .

$$\therefore \text{Req. no.} = p(m) = \sum_{i=1}^n S(m, i)$$

$$p(4) = \sum_{i=1}^3 S(4, i) = S(4, 1) + S(4, 2) + S(4, 3) \quad \rightarrow (1)$$

$$S(4, 1) = \frac{1}{1!} \sum_{k=0}^1 (-1)^k \binom{1}{1-k} (1-k)^4 = 1.$$

$$S(4, 2) = \frac{1}{2!} \sum_{k=0}^2 (-1)^k \binom{2}{2-k} (2-k)^4 = \frac{1}{2} [2^4 - 2 \times 1^4] = 7$$

$$\text{ii) } S(4, 3) = 6.$$

$$\therefore \textcircled{1} \Rightarrow \text{Req. no.} = 1 + 7 + 6 = 14.$$

14) Find the no. of equivalence relations that can be defined on a finite set A with  $|A|=6$ .

Soln:- Since  $|A|=m=6$ , a partition of A can have at most 6 cells. Treating the elements of A as objects & cells as con-

tainers, the no. of partitions having k cells is  $S(6, k)$ .

Since k varies from 1 to 6, the total no. of different partitions of A is

$$p(m) = \sum_{i=1}^n S(m, i)$$

$$\therefore p(6) = \sum_{i=1}^6 s(6, i)$$

$$p(6) = s(6, 1) + s(6, 2) + s(6, 3) + s(6, 4) + s(6, 5) + s(6, 6) \rightarrow (1)$$

we know that  $s(m, n) = \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m$

$$s(6, 1) = \frac{1}{1!} \sum_{k=0}^1 (-1)^k \binom{1}{1-k} (1-k)^6 = 1.$$

$$s(6, 2) = \frac{1}{2!} \sum_{k=0}^2 (-1)^k \binom{2}{2-k} (2-k)^6 = \frac{1}{2} [2C_2 \times 2^6 - 2C_1 \times 1^6] = 31$$

$$s(6, 3) = \frac{1}{3!} \sum_{k=0}^3 (-1)^k \binom{3}{3-k} (3-k)^6 = \frac{1}{6} [3C_3 \times 3^6 - 3C_2 \times 2^6 + 3C_1 \times 1^6] = 90$$

$$s(6, 4) = 65, \quad s(6, 5) = 15, \quad s(6, 6) = 1.$$

$\therefore$  No. of partitions of A is

$$(1) \Rightarrow p(6) = 203.$$

Since each partition of A corresponds to an equivalence rel<sup>n</sup> on A, it follows that if  $|A| = 6$ , then 203 equivalence relations can be defined on A.

15) Let  $A = \mathbb{R}$ ,  $B = \{x \mid x \text{ is real and } x \geq 0\}$ . IS the  $f^n f: A \rightarrow B$  defined by  $f(a) = a^2$  an onto  $f^n$ ? IS it a 1-1  $f^n$ ?

Sol<sup>n</sup>: - i) Take any  $b \in B$ , then  $b$  is non-negative real no.

Since  $f(a) = a^2 \Rightarrow b = a^2 \Rightarrow a = \pm \sqrt{b} \in A$  ( $\because A = \mathbb{R}$ ).

we see that, since  $f(a) = a^2$ ,  
 $f(\sqrt{b}) = (\sqrt{b})^2 = b$ .  
 $f(-\sqrt{b}) = (-\sqrt{b})^2 = b$ .

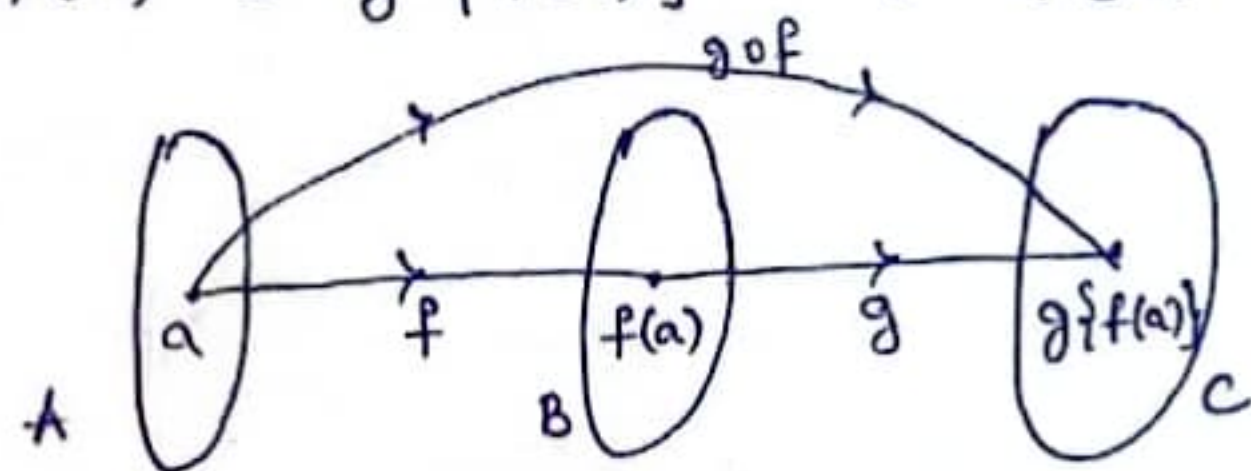
Thus  $\pm \sqrt{b}$  are the preimages of  $b$  under  $f$ .  
 Since 'b' is arbitrary element of B, every element in B has a preimage in A.  
 $\therefore f$  is onto  $f^n$ .

ii) Since  $b \in B$  has 2 preimages  $\pm \sqrt{b} \in A$  under  $f$ ,  
 $f$  is not one-to-one.

## Composition of functions :-

Consider 3 non-empty sets  $A, B, C$  and the fns  $f: A \rightarrow B$  and  $g: B \rightarrow C$ . The composition (or product) of these 2 functions is defined as  $g \circ f: A \rightarrow C$  with

$$(g \circ f)(a) = g\{f(a)\} \quad \forall a \in A.$$



Note :-  $\Rightarrow$  For  $f: A \rightarrow A$ ,  $f^1 = f$ ,  $f^2 = f \circ f$ ,  $f^3 = f \circ f^2$  & so on.  
Thus  $f^1 = f$ ,  $f^n = f \circ f^{n-1}$ .

$\rightarrow$  done in Problem(2)

2) Let  $f: A \rightarrow B$  and  $g: B \rightarrow C$ , then

- If  $f$  and  $g$  are one-to-one, then  $g \circ f$  is also one-to-one.
- If  $g \circ f$  is one-to-one, then  $f$  is one-to-one.
- If  $f$  and  $g$  are onto, then  $g \circ f$  is also onto.
- If  $g \circ f$  is onto, then  $g$  is onto.

3) The composition of 2 fns is not commutative  $\underline{\underline{=}} g \circ f \neq f \circ g$ .

### Problems :-

$\Rightarrow$  Let  $f: A \rightarrow B$ ,  $g: B \rightarrow C$  and  $h: C \rightarrow D$  be 3 functions.

$$\text{Then } (h \circ g) \circ f = h \circ (g \circ f).$$

Soln:- Since  $f: A \rightarrow B$ ,  $g: B \rightarrow C$  and  $h: C \rightarrow D$ , we have

both  $(h \circ g) \circ f$  and  $h \circ (g \circ f)$  are fns from  $A$  to  $D$ .

For any  $x \in A$ ,

$$\begin{aligned} [(h \circ g) \circ f](x) &= (h \circ g)[f(x)] = (h \circ g)(y) \quad \text{where } y = f(x) \\ &= h[g(y)] = h(z) \quad \text{where } z = g(y). \end{aligned}$$

$\hookrightarrow \textcircled{1}$

$$\begin{aligned} [h \circ (g \circ f)](x) &= h[(g \circ f)(x)] \\ &= h[g\{f(x)\}] \\ &= h[g(y)] = h(z) \rightarrow \textcircled{2} \end{aligned}$$

Thus  $[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x)$  for every  $x \in A$ .

$\therefore (h \circ g) \circ f = h \circ (g \circ f) \Rightarrow$  Compositions of 3 fns is associative.

- 2) Let  $f: A \rightarrow B$  and  $g: B \rightarrow C$ , Prove that
- If  $f$  and  $g$  are 1-1, then  $g \circ f$  is 1-1.
  - If  $g \circ f$  is 1-1, then  $f$  is 1-1.
  - If  $f$  and  $g$  are onto, then  $g \circ f$  is onto.
  - If  $g \circ f$  is onto, then  $g$  is onto.

Soln:- Let  $a_1, a_2 \in A$ .

i) Suppose  $(g \circ f)(a_1) = (g \circ f)(a_2)$ .

$$\Rightarrow g(f(a_1)) = g(f(a_2))$$

$$\Rightarrow g(b_1) = g(b_2) \quad [\because f: A \rightarrow B \Rightarrow b_1 = f(a_1) \text{ \& \& } b_2 = f(a_2)]$$

$$\Rightarrow b_1 = b_2 \quad (\because g \text{ is 1-1})$$

$$\Rightarrow f(a_1) = f(a_2)$$

$$\Rightarrow a_1 = a_2 \quad (\because f \text{ is 1-1})$$

Thus  $(g \circ f)(a_1) = (g \circ f)(a_2)$ , then  $a_1 = a_2$ .

$$\Rightarrow g \circ f \text{ is 1-1.}$$

ii) Let  $f(a_1) = f(a_2)$ .

$$\Rightarrow g(f(a_1)) = g(f(a_2))$$

$$\Rightarrow (g \circ f)(a_1) = (g \circ f)(a_2)$$

$$\Rightarrow a_1 = a_2 \quad (\because g \circ f \text{ is 1-1})$$

Thus if  $f(a_1) = f(a_2)$ , then  $a_1 = a_2$

$$\Rightarrow f \text{ is 1-1.}$$

iii)  $f$  is onto  $\Rightarrow \forall b \in B, \exists a \in A$  such that  $b = f(a) \rightarrow \textcircled{1}$

$g$  is onto  $\Rightarrow \forall c \in C, \exists b \in B$  such that  $c = g(b) \rightarrow \textcircled{2}$

from  $\textcircled{2}$ ,  $c = g(b) = g(f(a))$  (using  $\textcircled{1}$ )

$$c = (g \circ f)(a)$$

$\therefore \forall c \in C, \exists a \in A$  such that  $c = (g \circ f)(a) \Rightarrow g \circ f$  is onto.

iv)  $g \circ f$  is onto  $\Rightarrow \forall c \in C, \exists a \in A$  such that  $c = (g \circ f)(a)$ .

$$\Rightarrow c = g(f(a))$$

$$= g(b), \quad b = f(a) \in B.$$

$\therefore \forall c \in C, \exists b = f(a) \in B$  such that  $c = g(b)$

$\therefore g$  is onto.

3) Let  $A = \{1, 2, 3, 4\}$  and  $f: A \rightarrow A$  is a  $f^n$  defined by

$$f = \{(1, 2) (2, 2) (3, 1) (4, 3)\} \text{ . Find } f^2.$$

Soln:- Given:  $f(1) = 2, f(2) = 2, f(3) = 1, f(4) = 3$ .

$$\therefore f^2(1) = f \circ f(1) = f(f(1)) = f(2) = 2.$$

$$f^2(2) = f \circ f(2) = f(f(2)) = f(2) = 2.$$

$$f^2(3) = f \circ f(3) = f(f(3)) = f(1) = 2.$$

$$f^2(4) = f \circ f(4) = f(f(4)) = f(3) = 1.$$

$$\therefore f^2 = \{(1, 2) (2, 2) (3, 2) (4, 1)\}.$$

4) Consider the fun  $f$  and  $g$  defined by  $f(x) = x^3$  and  $g(x) = x^2 + 1$ ,  
 $\forall x \in \mathbb{R}$ , find  $g \circ f$ ,  $f \circ g$ ,  $f^2$  and  $g^2$ .

Soln:-  $(g \circ f)(x) = g\{f(x)\} = g(x^3) = (x^3)^2 + 1 = x^6 + 1.$

$$(f \circ g)(x) = f\{g(x)\} = f(x^2 + 1) = (x^2 + 1)^3 = (x^2)^3 + 1^3 + 3 \cdot x^2 \cdot 1(x^2 + 1) \\ = x^6 + 1 + 3x^4 + 3x^2$$

$$f^2(x) = (f \circ f)(x) = f\{f(x)\} = f(x^3) = (x^3)^3 = x^9$$

$$g^2(x) = (g \circ g)(x) = g\{g(x)\} = g(x^2 + 1) = (x^2 + 1)^2 + 1 = x^4 + 2 + 2x^2.$$

5) Let  $A = \{1, 2, 3, 4\}$  and  $B = \{a, b, c\}$  and  $C = \{x, y, z\}$  with

$f: A \rightarrow B$  and  $g: B \rightarrow C$  given by

$$f = \{(1, a) (2, a) (3, b) (4, c)\} \text{ and } g = \{(a, x) (b, y) (c, z)\} \text{ . Find } g \circ f.$$

Soln:- By data,

|            |     |            |
|------------|-----|------------|
| $f(1) = a$ | and | $g(a) = x$ |
| $f(2) = a$ |     | $g(b) = y$ |
| $f(3) = b$ |     | $g(c) = z$ |
| $f(4) = c$ |     |            |

Since  $f: A \rightarrow B$  and  $g: B \rightarrow C$ ,  $g \circ f: A \rightarrow C$ .

$$\therefore g \circ f(1) = g(f(1)) = g(a) = x.$$

$$g \circ f(2) = g(f(2)) = g(a) = x.$$

$$g \circ f(3) = g(f(3)) = g(b) = y.$$

$$g \circ f(4) = g(f(4)) = g(c) = z.$$

$$\text{Thus } g \circ f = \{(1, x) (2, x) (3, y) (4, z)\}.$$

P.T.O.

6) Let  $f, g, h$  be fns from  $\mathbb{Z}$  to  $\mathbb{Z}$  defined by

$$f(x) = x-1, \quad g(x) = 3x, \quad h(x) = \begin{cases} 0, & \text{if } x \text{ is even} \\ 1, & \text{if } x \text{ is odd.} \end{cases}$$

Determine  $(f \circ (g \circ h))(x)$  and  $((f \circ g) \circ h)(x)$  and verify that

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

Soln  $\rightarrow$  we have  $(g \circ h)(x) = g\{h(x)\} = 3h(x)$ .

$$\begin{aligned} \therefore (f \circ (g \circ h))(x) &= f\{(g \circ h)(x)\} \\ &= f\{g\{h(x)\}\} \\ &= f\{3h(x)\} = 3h(x) - 1. \end{aligned}$$

$$= \begin{cases} -1 & \text{if } x \text{ is even} \\ 2 & \text{if } x \text{ is odd.} \end{cases} \rightarrow \textcircled{1}$$

$$\boxed{\begin{aligned} ((f \circ g) \circ h)(x) &= (f \circ g)\{h(x)\} \\ &= f\{g\{h(x)\}\} \end{aligned}}$$

we have  $(f \circ g)(x) = f\{g(x)\} = g(x) - 1 = 3x - 1$ .

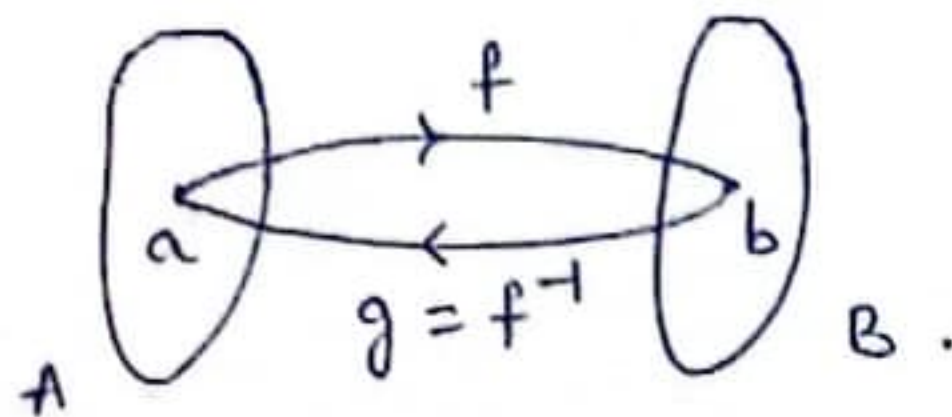
$$\begin{aligned} \therefore ((f \circ g) \circ h)(x) &= (f \circ g)\{h(x)\} \\ &= 3h(x) - 1 = \begin{cases} -1 & \text{if } x \text{ is even} \\ 2 & \text{if } x \text{ is odd.} \end{cases} \rightarrow \textcircled{2} \end{aligned}$$

from  $\textcircled{1}$  &  $\textcircled{2}$ ,

$$\begin{aligned} f \circ (g \circ h) &= (f \circ g) \circ h \\ &= \end{aligned}$$

Invertible functions :- A fn  $f: A \rightarrow B$  is said to be invertible if there exists a fn  $g: B \rightarrow A$  such that  $g \circ f = I_A$  and  $f \circ g = I_B$ , where  $I_A$  is the Identity fn on  $A$  and  $I_B$  is the Identity fn on  $B$ . Then  $g$  is called an inverse of  $f$  and we write  $g = f^{-1}$ .

$$\left[ \begin{array}{l} \text{since } f: A \rightarrow B, g: B \rightarrow A \\ g \circ f: A \rightarrow A \\ f \circ g: B \rightarrow B \end{array} \right.$$



Note :-

- 1) If a fn  $f: A \rightarrow B$  is invertible then it has a unique inverse. further, if  $f(a) = b$  then  $f^{-1}(b) = a$ .
- 2) If  $f$  is invertible, then  $f(a) = b$  and  $a = f^{-1}(b)$  are equivalent.
- 3) If  $f = \{ (a, b) \mid a \in A, b \in B \}$  is invertible, then  $f^{-1} = \{ (b, a) \mid b \in B, a \in A \}$  and conversely.
- 4) If  $f$  is invertible then  $f^{-1}$  is invertible and  $(f^{-1})^{-1} = f$ .
- 5) A fn  $f: A \rightarrow B$  is invertible iff it is one-to-one and onto.
- 6) If  $f: A \rightarrow B$  and  $g: B \rightarrow C$  are invertible fns then  $g \circ f: A \rightarrow C$  is also invertible and  $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ .
- 7) Let  $A$  and  $B$  be finite sets with  $|A| = |B|$  and  $f$  be a fn from  $A$  to  $B$ . Then the following statements are equivalent
  - i)  $f$  is one-to-one.
  - ii)  $f$  is onto.
  - iii)  $f$  is invertible.

P.T.O

### Problems :-

1) Suppose  $f: A \rightarrow B$  is invertible, then P.T  
Prove that  $f$  has unique inverse. further, if  $f(a) = b$  then  
 $a = f^{-1}(b)$ .

Soln:- Given  $f$  is invertible.

Let  $g$  be inverse of  $f$ . I.e.  $g = f^{-1}$ .

$$\therefore g \circ f = I_A \quad \text{and} \quad f \circ g = I_B.$$

Let us assume there is one more inverse for  $f$ , say  $h$ .

$$\therefore h \circ f = I_A \quad \text{and} \quad f \circ h = I_B.$$

$$\text{consider } h \circ (f \circ g) = (h \circ f) \circ g.$$

$$h \circ I_B = I_A \circ g.$$

$$\therefore h = g \quad \#$$

$\therefore f$  has unique inverse.

Further,  $f(a) = b$ .

$$\Rightarrow g(f(a)) = g(b)$$

$$\Rightarrow g \circ f(a) = g(b)$$

$$\Rightarrow I_A(a) = g(b)$$

$$\Rightarrow a = g(b)$$

$$\Rightarrow a = f^{-1}(b) \quad (\because g = f^{-1})$$

2) Q.P  $f: A \rightarrow B$  is invertible iff it is one-to-one and onto.

Soln:- Let  $f: A \rightarrow B$  be a invertible fn, then there exists a unique fn  $g: B \rightarrow A$  such that  $g \circ f = I_A$  and  $f \circ g = I_B$ .

T.P.T  $f$  is one-to-one.

$$\text{Let } f(a_1) = f(a_2)$$

$$\Rightarrow g(f(a_1)) = g(f(a_2))$$

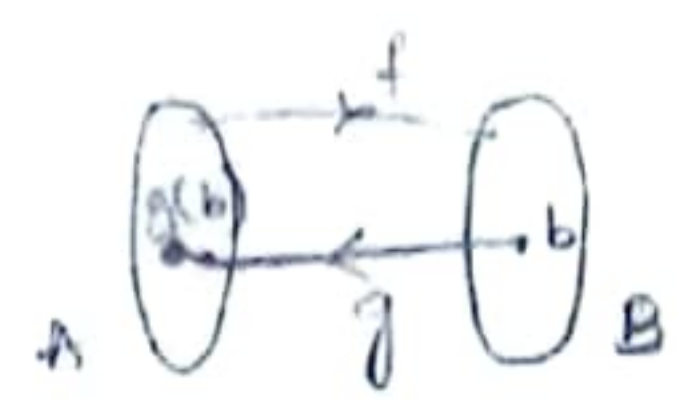
$$\Rightarrow g \circ f(a_1) = g \circ f(a_2)$$

$$\Rightarrow I_A(a_1) = I_A(a_2)$$

$$\Rightarrow a_1 = a_2$$

$\therefore f$  is 1-1.

Th  $f$  is onto.



Take any  $b \in B$ , then  $g(b) \in A$ .

$$\text{and } b = I_B(b) = (f \circ g)(b) = f\{g(b)\}$$

Thus  $b$  is the image of an element  $g(b) \in A$  under  $f$ .

$\therefore f$  is onto.

Conversely, Suppose  $f$  is 1-1 and onto.  $\therefore f$  is bijective.

$\therefore$  for every  $b \in B$  there exists a unique  $a \in A$  such that

$$f(a) = b.$$

Consider a fn  $g: B \rightarrow A$  defined by  $g(b) = a$ .

$$\text{then } (f \circ g)(b) = f\{g(b)\} = f(a) = b = I_B(b).$$

$$(g \circ f)(a) = g\{f(a)\} = g(b) = a = I_A(a).$$

$\therefore f$  is Invertible with  $g$  as the Inverse.

3) If  $f: A \rightarrow B$  and  $g: B \rightarrow C$  are invertible fns, then  $g \circ f: A \rightarrow C$  is also invertible and  $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ . Prove that

Soln:- Suppose  $f$  and  $g$  are invertible.

$\therefore f$  and  $g$  are 1-1 and onto.

$\therefore g \circ f$  is also 1-1 and onto.

Hence  $g \circ f$  is Invertible.

Next, we have  $f: A \rightarrow B$ ,  $g: B \rightarrow C$ ,  $g \circ f: A \rightarrow C$ .

$\therefore f^{-1}: B \rightarrow A$ ,  $g^{-1}: C \rightarrow B$ , and let  $h = f^{-1} \circ g^{-1}: C \rightarrow A$ .

$$\begin{aligned} \text{consider } h \circ (g \circ f) &= (f^{-1} \circ g^{-1}) \circ (g \circ f) \\ &= f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ I_B \circ f = f^{-1} \circ f = I_A \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} \text{consider } (g \circ f) \circ h &= (g \circ f) \circ (f^{-1} \circ g^{-1}) \\ &= g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ I_A \circ g^{-1} = g \circ g^{-1} = I_C \quad \text{--- (2)} \end{aligned}$$

from (1) & (2),  $h$  is the inverse of  $g \circ f$ .

$$\underline{\underline{h = (g \circ f)^{-1}}}$$

$$\text{Thus } (g \circ f)^{-1} = h = f^{-1} \circ g^{-1}.$$

4) Let  $A = \{1, 2, 3, 4\}$  and  $f$  and  $g$  be fns from  $A$  to  $A$  given by  $f = \{(1, 4)(2, 1)(3, 2)(4, 3)\}$  and  $g = \{(1, 2)(2, 3)(3, 4)(4, 1)\}$ .

Prove that  $f$  and  $g$  are inverses of each other.

Soln:- By data,  $f(1) = 4, f(2) = 1, f(3) = 2, f(4) = 3$   
 $g(1) = 2, g(2) = 3, g(3) = 4, g(4) = 1$ .

we have

$$\begin{aligned}(g \circ f)(1) &= g\{f(1)\} = g(4) = 1 = I_A(1) \\ (g \circ f)(2) &= g\{f(2)\} = g(1) = 2 = I_A(2) \\ (g \circ f)(3) &= g\{f(3)\} = g(2) = 3 = I_A(3) \\ (g \circ f)(4) &= g\{f(4)\} = g(3) = 4 = I_A(4)\end{aligned}$$

and

$$\begin{aligned}(f \circ g)(1) &= f\{g(1)\} = f(2) = 1 = I_A(1) \\ (f \circ g)(2) &= f\{g(2)\} = f(3) = 2 = I_A(2) \\ (f \circ g)(3) &= f\{g(3)\} = f(4) = 3 = I_A(3) \\ (f \circ g)(4) &= f\{g(4)\} = f(1) = 4 = I_A(4)\end{aligned}$$

Thus  $\forall x \in A, (g \circ f)(x) = I_A(x)$  and  $(f \circ g)(x) = I_A(x)$ .

$\Rightarrow g$  is an inverse of  $f$  and  $f$  is an inverse of  $g$ .

$\therefore$   $f$  and  $g$  are inverses of each other.

5) Let  $A = B = C = \mathbb{R}$  and  $f: A \rightarrow B$  and  $g: B \rightarrow C$  be defined by  
Q.P  $f(a) = 2a + 1$  and  $g(b) = \frac{1}{3}b, \forall a \in A, \forall b \in B$ .

compute  $g \circ f$  and show that  $g \circ f$  is invertible. find  $(g \circ f)^{-1}$ .

Soln:- we have  $(g \circ f)(a) = g\{f(a)\} = g(2a + 1) = \frac{1}{3}(2a + 1)$ .

Thus  $g \circ f: A \rightarrow C$  is defined by  $(g \circ f)(a) = \frac{1}{3}(2a + 1)$

we have  $b = f(a) = 2a + 1 \Rightarrow a = \frac{b-1}{2}$ .

$$\begin{aligned}f(a_1) &= f(a_2) \\ \Rightarrow 2a_1 + 1 &= 2a_2 + 1 \\ \Rightarrow 2a_1 &= 2a_2 \Rightarrow a_1 &= a_2 \\ \Rightarrow f &\text{ is 1-1}\end{aligned}$$

$\therefore a = f^{-1}(b) = \frac{b-1}{2} \Rightarrow f$  is onto  $\Rightarrow f$  is invertible.

$$\begin{aligned}g(b_1) &= g(b_2) \\ \Rightarrow \frac{b_1}{3} &= \frac{b_2}{3} \\ \Rightarrow b_1 &= b_2 \\ \Rightarrow g &\text{ is 1-1}\end{aligned}$$

Also  $c = g(b) = \frac{1}{3}b \Rightarrow b = 3c$ .

$\therefore b = g^{-1}(c) = 3c \Rightarrow g$  is onto  $\Rightarrow g$  is invertible.

Thus  $g \circ f$  is invertible and its inverse is given by

$$(g \circ f)^{-1}(c) = (f^{-1} \circ g^{-1})(c) = f^{-1}\{g^{-1}(c)\} = f^{-1}(3c) = \frac{3c-1}{2}$$

$$\begin{aligned}
 \therefore (g \circ f)^{-1}(c) &= (f^{-1} \circ g^{-1})(c) \\
 &= f^{-1}\{g^{-1}(c)\} = f^{-1}\{3c\} \\
 &= \frac{1}{2}(3c-1).
 \end{aligned}$$

6) Let  $A = B = \mathbb{R}$ , the set of all real no.'s, and the fns  
Q.P  $f: A \rightarrow B$  and  $g: B \rightarrow A$  be defined by

$$f(x) = 2x^3 - 1, \forall x \in A; \quad g(y) = \left\{ \frac{1}{2}(y+1) \right\}^{1/3}, \forall y \in B.$$

Show that each of  $f$  and  $g$  is the inverse of <sup>the</sup> other.

Soln:- for any  $x \in A$ ,

$$\begin{aligned}
 (g \circ f)(x) &= g\{f(x)\} = g(y) \quad \text{where } y = f(x) \\
 &= \left\{ \frac{1}{2}(y+1) \right\}^{1/3} \\
 &= \left\{ \frac{1}{2}(2x^3 - 1 + 1) \right\}^{1/3} \quad (\because y = f(x) = 2x^3 - 1) \\
 &= x.
 \end{aligned}$$

$$\therefore g \circ f = I_A.$$

for any  $y \in B$ ,

$$\begin{aligned}
 (f \circ g)(y) &= f\{g(y)\} \\
 &= f(z) \quad \text{where } z = g(y) \\
 &= 2z^3 - 1 \\
 &= 2(g(y))^3 - 1 = 2\left(\left\{ \frac{1}{2}(y+1) \right\}^{1/3}\right)^3 - 1 \\
 &= 2\left\{ \frac{1}{2}(y+1) \right\} - 1 = y + 1 - 1 = y.
 \end{aligned}$$

$$\therefore f \circ g = I_B.$$

$\therefore$  each of  $f$  and  $g$  is an invertible fn.

$\Rightarrow f$  and  $g$  are inverses of each other.

## THE PIGEONHOLE PRINCIPLE

Statement:- If 'm' pigeons occupy 'n' pigeonholes and if  $m > n$ , then two or more pigeons occupy the same pigeonhole.

(OR)

If 'm' pigeons occupy 'n' pigeonholes where  $m > n$ , then at least one pigeonhole must contain two or more pigeons in it.

Ex:- If 8 children are born in the same week, then two or more children are born on the same day of the week.

Generalised pigeon hole Principle:-

If 'm' pigeons occupy 'n' pigeonholes and  $m > n$ , then at least one pigeonhole must contain  $p+1$  or more pigeons in it,

where  $p = \left\lfloor \frac{(m-1)}{n} \right\rfloor$

Problems:-

➤ ABC is an equilateral  $\Delta^e$  whose sides are of length 1 cm each. If we select 5 points inside the  $\Delta^e$ , prove that at least 2 of these points are such that the distance b/w them is less than  $\frac{1}{2}$  cm.

Soln:-



Consider the equilateral  $\Delta^e$  ABC whose sides are of length 1 cm each. Consider the  $\Delta^e$  DEF formed by the midpoints of the sides

AB, AC and BC resp (see fig). This divides  $\Delta^e$  ABC into 4 small equilateral  $\Delta^e$ s, whose length of each side is  $\frac{1}{2}$  cm.

Let us treat 5 points as pigeons and 4 small equilateral  $\Delta^e$ s as pigeon holes, then by pigeon hole principle, at least one small  $\Delta^e$  contains 2 or more points and the distance b/w such points is less than  $\frac{1}{2}$  cm.

2) A bag contains 12 pair of socks (each pair in different color).  
If a person draws the socks one by one at random,  
determine at most how many draws are required to get  
at least one pair of matched socks.

Soln:- Let  $n$  be the no. of draws.

For  $n \leq 12$ , it is possible that the socks drawn are of different colors  $\because$  there are 12 colors.

For  $n = 13$ , all socks cannot have different colors - at least two must have same color.

Let us treat 13 as the no. of pigeons and 12 colors as 12 pigeonholes.  
 $\therefore$  at most 13 draws are required to have at least 1 pair of socks of the same color.

3) If 5 colors are used to paint 26 doors, prove that at least 6 doors will have same color.

Soln:- Consider 26 doors as pigeons and 5 colors as pigeon holes. By generalised pigeon hole principle, at least 1 color must be assigned to  $p+1$  or more doors.

$$\text{i.e. } p+1 = \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{26-1}{5} \right\rfloor + 1 = 5+1 = 6.$$

Do 5th one here.

4) How many persons must be chosen in order that at least five of them will have birthdays in the same calendar month?

Soln:- Let ' $m$ ' be the no. of persons. Number of months

over which the birthdays are distributed is  $n = 12$ .

The least no. of persons having birthday in the same month

$$\text{is } 5 = p+1.$$

$$\text{i.e. } p+1 = 5 \Rightarrow \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 5 \Rightarrow \left\lfloor \frac{m-1}{12} \right\rfloor = 4$$

$$\Rightarrow m-1 = 48 \Rightarrow \boxed{m=49}$$

∴ No. of persons is 49 (at the least).

5) Prove that if 30 dictionaries in a library contain a total of 61,327 pages, then at least one of the dictionaries must have at least 2045 pages.

Soln: Consider 61,327 pages as pigeons  $\underline{u}$   $m = 61,327$  and 30 dictionaries as pigeon holes  $\underline{u}$   $n = 30$ .

By using the generalised pigeonhole principle, at least 1 dictionary must contain  $p+1$  or more pages.

$$\underline{u} \quad p+1 = \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{61,327-1}{30} \right\rfloor + 1 = \lfloor 2044.2 \rfloor + 1 \\ = 2044 + 1 \\ = 2045.$$

This proves the required result.

6) If any  $n+1$  numbers are chosen from 1 to  $2n$ , then show that at least one pair add to  $2n+1$ .

Soln: - let us consider the following sets:

$$A_1 = \{1, 2n\}, A_2 = \{2, 2n-1\}, A_3 = \{3, 2n-2\} \dots$$

$$A_{n-1} = \{n-1, n+2\}, A_n = \{n, n+1\}.$$

These are the only sets containing 2 no's from 1 to  $2n$  whose sum is  $2n+1$ .

Since there are only  $n$  sets, two of the  $n+1$  chosen numbers belong to same set (by pigeon hole principle) and sum of these 2 no's =  $2n+1$ .

P.T.O.

7) Prove that if 101 integers are selected from the set  $S = \{1, 2, 3, \dots, 200\}$ , then at least 2 of these are such that one divides the other.

Soln:- we have  $S = \{1, 2, 3, \dots, 200\}$ .

Let  $X = \{1, 3, 5, \dots, 199\} \Rightarrow |X| = 100$ .

Any element  $n$  in the set  $S$  can be written as

$n = 2^k \times x$  where  $k$  is an integer  $\geq 0$  and  $x \in X$ .

Ex:-  $1 = 2^0 \times 1$

$2 = 2^1 \times 1$

$3 = 2^0 \times 3$

$4 = 2^2 \times 1$  etc

Considering 101 elements as pigeons and elements in  $X$  as pigeonholes, by pigeonhole principle, at least 2 elements out of 101 must be related to same  $x \in X$ .

Let  $a$  and  $b$  be such elements, then  $a = 2^{k_1} x$ ,  $b = 2^{k_2} x$ , where  $k_1$  and  $k_2$  are integers  $\geq 0$ .

Thus if  $k_1 \leq k_2$  then  $a$  divides  $b$ .

if  $k_1 > k_2$  then  $b$  divides  $a$ .  
(i.e.  $k_2 < k_1$ )

8) Shirts numbered consecutively from 1 to 20 are worn by 20 students of a class. When any 3 of these students are chosen to be a debating team from the class, the sum of their shirt numbers is used as the code no. of the team. Show that if any 8 of the 20 are selected, then <sup>from</sup> these 8 we may form at least 2 different teams having the same code no.

Soln:- No. of ways of choosing 3 ~~team of~~ students ~~out of~~ 8 is  ${}^8C_3 = 56$  - P.T.O.

This is the no. of teams possible.

Smallest code no. is  $1+2+3 = 6$ .

Soln:- From the 8 of the 20 students selected, the no. of teams of 3 students that can be formed is  ${}^8C_3 = 56$ .

Smallest possible code no. is  $1+2+3 = 6$ .

Largest " " " "  $18+19+20 = 57$ .

∴ code no.'s vary from 6 to 57 (inclusive) and there are 52 in no.

Let us consider no. of teams as pigeons i.e.  $m = 56$ .

and the no. of codes as pigeonholes i.e.  $n = 52$ .

∴ By pigeon hole principle, at least 2 different teams will have the same code no, i.e. at least 1 code must be assigned to 2 or more teams.